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Multimodal pooled Perturb-CITE-seq screens in patient models define mechanisms of cancer immune evasion

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Resistance to immune checkpoint inhibitors (ICIs) is a key challenge in cancer therapy. To elucidate underlying mechanisms, we developed Perturb-CITE-sequencing (Perturb-CITE-seq), enabling pooled clustered regularly interspaced short palindromic repeat (CRISPR)–Cas9 perturbations with single-cell transcriptome and protein readouts. In patient-derived melanoma cells and autologous tumor-infiltrating lymphocyte (TIL) co-cultures, we profiled transcriptomes and 20 proteins in ~218,000 cells under ~750 perturbations associated with cancer cell-intrinsic ICI resistance (ICR). We recover known mechanisms of resistance, including defects in the interferon-γ (IFN-γ)–JAK/STAT and antigen-presentation pathways in RNA, protein and perturbation space, and new ones, including loss/downregulation of *CD58*. Loss of *CD58* conferred immune evasion in multiple co-culture models and was downregulated in tumors of melanoma patients with ICR. CD58 protein expression was not induced by IFN-γ signaling, and *CD58* loss conferred immune evasion without compromising major histocompatibility complex (MHC) expression, suggesting that it acts orthogonally to known mechanisms of ICR. This work provides a framework for the deciphering of complex mechanisms by large-scale perturbation screens with multimodal, single-cell readouts, and discovers potentially clinically relevant mechanisms of immune evasion.

Molecular circuits form the basic driving force from genotype to phenotype in cells, tissues and entire organisms. Circuits in cells process diverse signals, make appropriate decisions and orchestrate physiological responses to these signals. Diseases arise from circuit malfunctions: one or more components is missing or defective or a key module is over- or underactivated. One key approach to charting circuits and understanding their function is pooled perturbation screens. Perturb-sequencing (Perturb-seq)[1–4] is a recently developed method that combines pooled genetic perturbation screens with single-cell RNA sequencing (scRNA-seq), enabling the perturbation to be read as an RNA barcode along with the full transcriptional profile of the cell as a rich readout. However, many relevant phenotypes are functionally best understood at the protein level and are reflected differently at the RNA and protein levels, thus requiring expansion of previous Perturb-seq schemes. A proof-of-concept study using expanded CRISPR-compatible cellular indexing of transcriptomes and epitopes by sequencing (ECCITE-seq)[5] demonstrated simultaneous detection of protein, RNA and guide RNAs (gRNAs) in individual cells and was applied at a small scale without the analytical framework required for large-scale screens.

Multimodal genetic screens provide a powerful opportunity to enhance our understanding of ICR. Antibodies targeting CTLA-4 or the PD-1–PD-L1 axis release tumor-mediated immune inhibition and enable T-cell-mediated killing of tumor cells[6]. While ICIs produce durable responses in some patients, most either do not benefit or develop resistance over time. Understanding mechanisms of resistance is therefore a key challenge and opportunity. Clinically relevant mechanisms include mutations in beta-2-microglobulin (*B2M*) resulting in loss of MHC Class I presentation, downregulation of the antigen-presentation machinery and mutations in the IFN-γ–JAK/STAT pathway that impair response to T-cell-mediated antitumor activity[7–9]. However, known genetic variants account for only a minority of cases. We previously identified an ICR signature

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predictive of intrinsic resistance to ICIs in a subset of malignant cells in patients with metastatic melanoma by using scRNA-seq[9]. CRISPR–Cas9 knockout (KO) screens in murine models or engineered human cell lines testing tumor–T-cell interactions identified putative new mechanisms of immune evasion involving chromatin regulators[10], TNF-α signaling[11], *SOX4* (ref. [12]), *PTPN2* (ref. [13]) and *APLNR*[10], yet their clinical relevance is unclear. Recent advances in cellular models allow us to study co-cultures of malignant cells derived directly from patient tumors along with their ex vivo expanded autologous TILs[14], and provide an excellent screening platform to study the effects of cancer immunotherapy[14–17].

Here, we develop Perturb-CITE-seq, an extension of Perturb-seq that combines scRNA-seq profiling and epitope sequencing (CITE-seq)[18] of single-cell surface proteins under perturbations[5]. We performed pooled Perturb-CITE-seq screens of ICR program genes in a patient-derived tumor–TIL co-culture model, targeting 248 genes of the ICR signature (744 targeting guides) and profiling single-cell transcriptomes and 20 surface proteins in >218,000 cells. We developed an integrated analytical framework and recovered the known landscape of clinically relevant immune resistance mutations, along with new ones. Among others, we find that loss or downregulation of *CD58* confers immune evasion to T-cell- (and, in part, to natural killer (NK) cell)-mediated killing, and we validate this finding in both additional patient models and clinical scRNA-seq data of patients with ICR. Our work provides a broadly applicable experimental and analytical framework for Perturb-CITE-seq in patient models and identifies mechanisms of immune evasion that are in part orthogonal to previously described mechanisms in patients.

Results

Patient-derived co-cultures for pooled perturbation screens. For systematic and functional evaluation of the contribution of ICR signature genes to T-cell-mediated killing resistance, we designed two types of CRISPR–Cas9 loss of function (KO) screens using a human tumor immune co-culture model[9] (Fig. 1a): a viability screen, to determine the impact of perturbation on T-cell-mediated killing (Fig. 1h), and a Perturb-CITE-seq screen, to decipher the underlying circuitry (Fig. 1i,j).

We established autologous co-culture models of melanoma cell lines and ex vivo expanded TILs from multiple patients (Fig. 1a; Methods). The expanded TILs consisted of either exclusively CD8$^+$ T cells (Extended Data Fig. 1b,d) or a mixed population of CD8$^+$ and CD4$^+$ T cells (Extended Data Fig. 1c). Using a miniaturized optical platform (Extended Data Fig. 1a; Methods) and improved methods of T-cell activation to reduce bystander killing (Extended Data Fig. 1b–h; Methods), we found that co-cultures of T cells and cancer cells resulted in dose- (TIL/cancer cell ratio) and time-dependent cancer cell killing (Fig. 1b,d,f). Tumor cell lysis was highly reproducible and specific to T-cell-receptor (TCR)–MHC Class I interactions (Fig. 1c,e,g and Extended Data Fig. 1h).

Next, for both viability and Perturb-CITE-seq pooled screens, we established a pooled library of 744 single-guide RNAs targeting 248 genes with putative roles in immunotherapy resistance (ICR library, Supplementary Table 1; Methods). We transduced patient-derived melanoma cells stably expressing active Cas9 (Extended Data Fig. 2a) with the ICR library at a multiplicity of infection of 0.1 to achieve a high rate of single-guide transductions with ~1,000 cells per guide (Extended Data Fig. 2b–d), cultured the cells for 14 days and then co-cultured them with a range of TIL doses (1:1, 2:1 or 4:1), treated them with IFN-γ (no co-culture) or maintained them in culture medium alone (control). Survivor cells of each condition were collected after 48 h of co-culture for gDNA isolation, sequencing and identification of enriched perturbations (Fig. 1h; Methods).

To develop Perturb-CITE-seq, we combined 3′ droplet-based scRNA with extracellular protein detection (CITE-seq) and single-cell gRNA detection[1–3]. We expressed a single gRNA (sgRNA) on a polyadenylated transcript using a modified CRISPR droplet-sequencing (CROP-seq) vector (Methods) and performed a targeted 'dial-out' amplification generating a robust dictionary linking gRNA identities to single-cell transcriptional and protein profiles. Notably, previously described methods used to detect gRNAs are compatible with Perturb-CITE-seq, including ECCITE-seq and direct capture of gRNAs by feature barcoding[1,2,5].

For the Perturb-CITE-seq screen, we collected surviving cells from each of the three conditions (Fig. 1j) with a representation of ~100 cells per perturbation[1]. We detected an average of 16 antibodies per cell (from a total of 24 antibodies including IgG controls) with an average of 45.2 unique molecular identifiers (UMIs) per antibody, and 89.5% of all cells with both an assigned sgRNA and multiple detected antibodies. The detection sensitivity for gRNA, protein and messenger RNA was similar to that previously described for Perturb-seq[1–3] and CITE-seq in separate experiments[18].

Pooled screens identify drivers of immune evasion. To identify 'essential' genes independent of T-cell-mediated killing in this robust screen (Extended Data Fig. 3a–c), we identified sgRNAs that were depleted by days 7 and 14 before any treatment or co-culture (Fig. 2a, Extended Data Fig. 3b–d and Supplementary Table 2); these included expected candidates such as *MYC*. On day 14, we treated cells with either IFN-γ or control medium for 16 h followed by either co-culture for 48 h at different TIL/tumor cell ratios or medium only (Fig. 1h). The screens were highly reproducible across triplicate samples (Fig. 2b,c and Extended Data Fig. 3b,c), and co-culture showed dose-dependent lysis of cancer cells with 30.96, 58.35 and 75.0% killing at ratios of 1:1, 2:1 and 4:1, respectively (Extended Data Fig. 3a). As expected, perturbations of *B2M*, *HLA-A*, *JAK1*, *JAK2*, *STAT1*, *IFNGR1* and *IFNGR2* (Fig. 2d,e, Extended Data Fig. 3e,f and Supplementary Table 2) conferred immune evasion, consistent with clinical mechanisms of resistance to immunotherapy[7–9].

Other enriched perturbations included *CD47*, *CD58* and *CDH19* (Fig. 2d,e). CD47 is a 'don't eat me' signal on tumor cells that interacts with SIRPa on phagocytic cells; blockade of the CD47–SIRPa axis improves tumor control in vivo[19]. Recent studies revealed that activated T cells also express SIRPa and that interaction with CD47 results in increased T-cell activation and cytokine production[20], which may impact the potential clinical benefit of CD47 blockade[21]. CD58 (also known as LFA3) is an understudied adhesion molecule, typically expressed on antigen-presenting cells, that binds CD2 on T cells and NK cells[22]. The role of CD58 in cancer is poorly understood. Notably, there is no known mouse homolog of *CD58*, emphasizing the value of performing such screens in human models[10]. Overall, the viability screen recovered known immune evasion regulators in a dose-dependent manner and identified several new candidates. We next explored our Perturb-CITE-seq data to organize these candidates according to pathway, along with the molecular processes they regulate.

Multimodal, single-cell readouts of immune evasion. Across all Perturb-CITE-seq screens, we analyzed 218,331 high-quality scRNA and protein profiles spanning the co-culture (73,114 cells), IFN-γ (87,590 cells) and control (57,627 cells) conditions (Fig. 3a,e and Extended Data Fig. 4a–d; Methods). We removed 805 contaminating T cells using unsupervised clustering (Extended Data Fig. 4a,b; Methods). We first focused on RNA and protein profiles in the context of the various conditions (irrespective of perturbations). Embedding of cells by either protein (Fig. 3a) or RNA (Fig. 3e) profiles alone were separated by treatment conditions. Within each condition, both the cell cycle (G1/0, S, G2/M) (Fig. 3f) and complexity (Extended Data Fig. 4c) impacted RNA profiles, revealing important covariates addressed in subsequent analyses (Extended Data Fig. 4d–h). At both the protein and RNA level, we identified

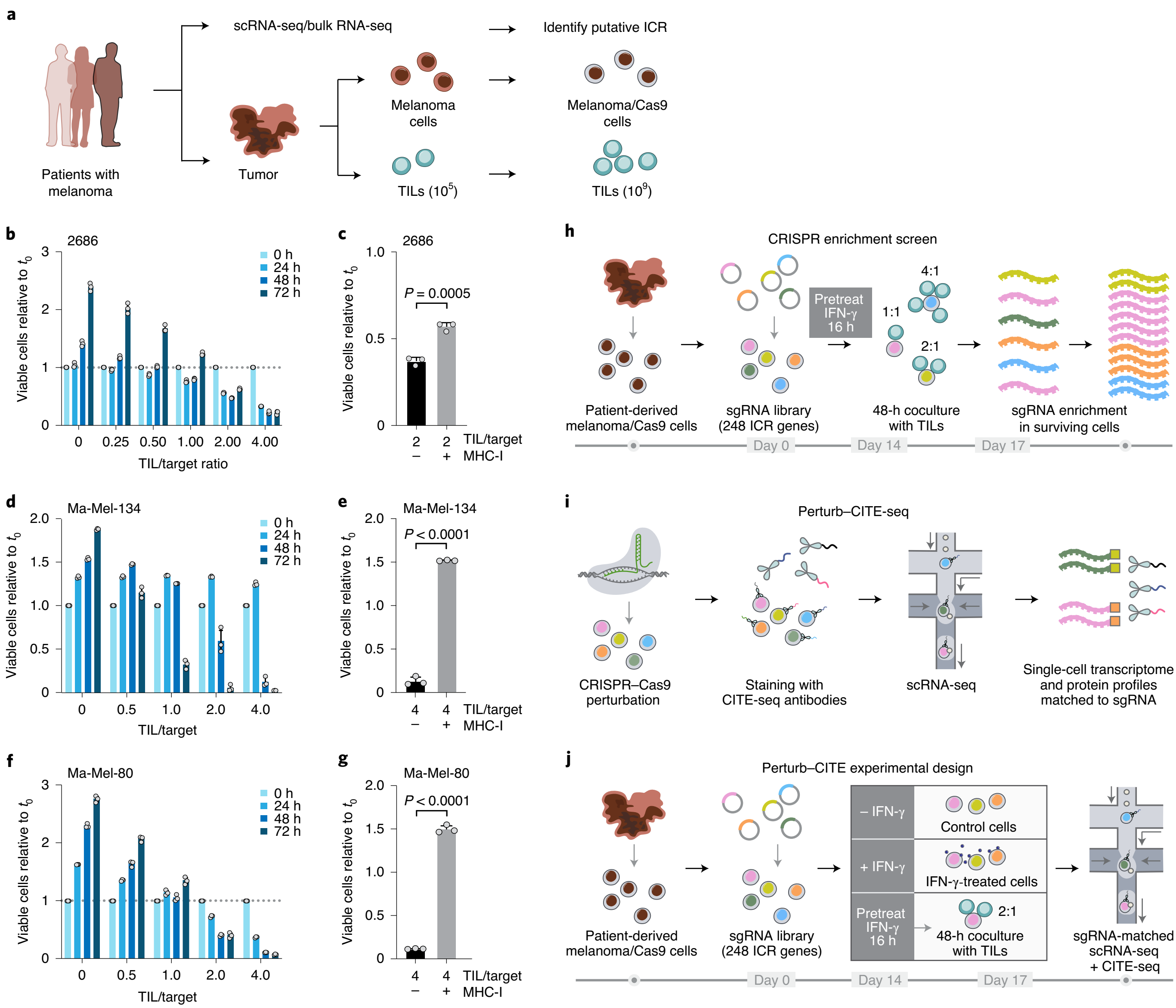


Fig. 1 | Use of Perturb-CITE-seq to study tumor-intrinsic mechanisms of T-cell resistance using patient-derived co-culture models. a, Patient-derived cell models and programs. Resected melanomas were profiled by bulk RNA- and scRNA-seq to identify an ICR[9]. Melanoma cells and TILs were grown from patients' tumors; melanoma cells were transduced to express Cas9 while TILs were expanded to yield sufficient numbers for screening (Methods). **b–g**, Optimization and validation of co-culture system for TIL-mediated killing of melanoma cells. **b**,**d**,**f**, Time- and dose-dependent killing of melanoma target cells (2686, **b**; Ma-Mel-134, **d**; Ma-Mel-80, **f**) by autologous TILs in three patient-derived co-culture models. Ratio of viable cancer cells (*y* axis, relative to t_0) at different TIL/cancer cell ratios (*x* axis) at different time points (color legend) after pretreatment of target cells with 1 ng ml^{-1} IFN-γ for 16 h using TILs without previous restimulation in each patient-derived co-culture model. The experiment was performed in triplicate in each of two independent experiments. **c**,**e**,**g**, Killing of target cells (2686, **c**; Ma-Mel-134, **e**; Ma-Mel-80, **g**) is dependent on MHC-I. Ratio of viable cancer cells (*y* axis, relative to t_0) after 48 h in a 4/1 TIL/cancer cell co-culture of cancer cells pretreated with 1 ng ml^{-1} IFN-γ for 16 h and TILs without previous restimulation, in the absence or presence of MHC-I-blocking antibodies (*x* axis). Two-sided *t*-test. Error bars, mean ± s.d. The experiment was performed in triplicate in each of two independent experiments. **h**, Viability screen design. **i**, Perturb-CITE-seq approach. **j**, Perturb-CITE-seq screens used to characterize regulators of melanoma immune evasion.

expected responses to T-cell-mediated killing (Supplementary Table 3; Methods).

At the protein level, we observed increases in MHC proteins and PD-L1 (*CD274*) in IFN-γ-treated versus control cells, a global increase in MHC proteins in co-culture versus control cells (Fig. 3b and Supplementary Table 4; Methods) and, consistent with previous reports[12], induction of *CD49f*, which encodes an integrin associated with epithelial-to-mesenchymal transition. Conversely, there was strong downregulation of several proteins with potential roles in modifying the response to immunotherapies exclusively in co-culture conditions, including CXCR4 (*CD184*), c-KIT (*CD117*) and KDR (*CD309*)[23,24] (Fig. 3b). In particular, levels of proteins CD58, CD47 and IFNGR1 (*CD119*) were reduced in co-culture, concordant with enrichment of their genetic KO in the viability screen (Figs. 2d,e and 3b). Comparison of RNA profiles between conditions also highlighted genes involved in antigen presentation, chemokines and immune modulators (Fig. 3g,h and Supplementary Table 4), yet only certain protein-level differences between treatments were observed at the level of the individual corresponding transcript (Fig. 3c), highlighting the importance of simultaneous

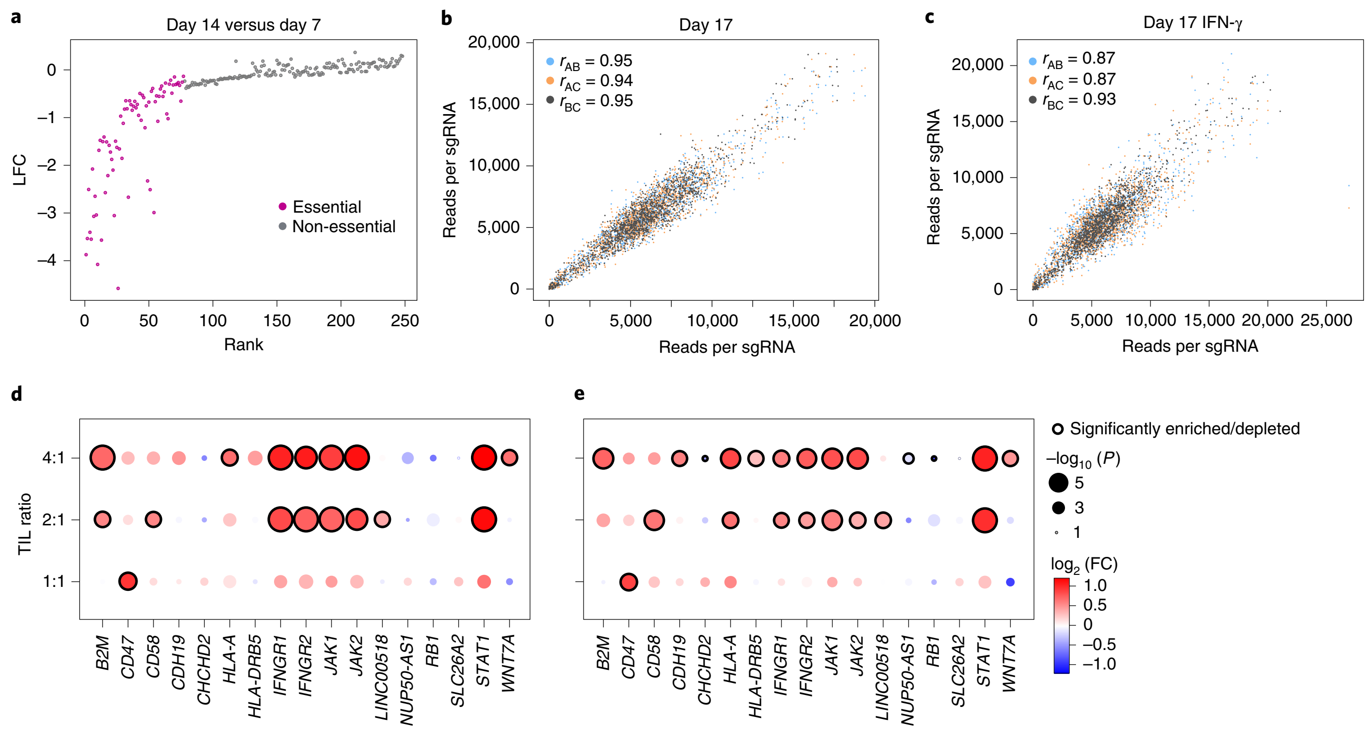


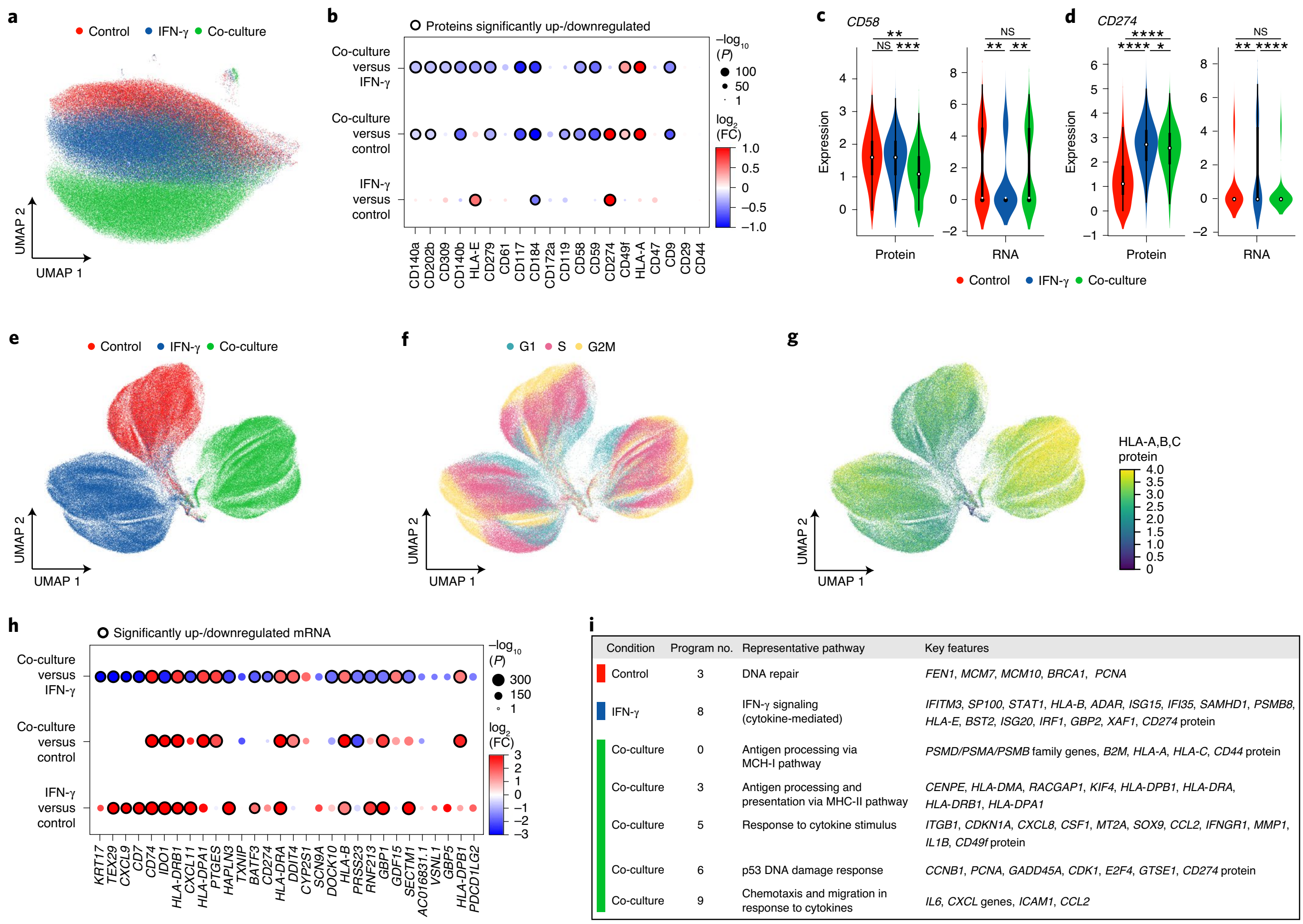
Fig. 2 | Identification of genes for evasion of TIL-mediated killing by CRISPR–Cas9 viability screen in patient-derived models exposed to increasing immune pressures. a, Identification of essential genes unrelated to immune pressure. Change in abundance (negative $\log_2$ fold-change (LFC), *y* axis) of each sgRNA (dot) at day 14 versus day 7 following lentivirus transduction, with guides ranked (*x* axis) by LFC value. Pink: called essential genes (Methods). **b**,**c**, High reproducibility of screen across triplicates. Number of reads detected (*x*, *y* axes) for each sgRNA (dots) when comparing each pair within triplicate experiments (color legend) in either control cells (**b**) or IFN-γ-treated cells (**c**) on day 17. Pearson (*r*) correlation coefficients are noted in the legend. **d**,**e**, Co-culture screen highlighting the role of the IFN-γ–JAK/STAT pathway and additional mechanisms. LFC (dot color) and significance ($-\log_{10}(P)$ and dot size (and significantly enriched/depleted perturbations) encircled with black border) (Methods) of sgRNA gene perturbations (columns) that are differentially enriched in tumor cells co-cultured with different doses of TILs (rows) on day 17 compared to control cells (**d**) or IFN-γ-treated cells (**e**).

RNA and protein detection. Overall, our results suggest that genes functionally impacting susceptibility to T-cell-mediated killing (based on genetic perturbation in the viability screen) are concordantly regulated at the protein and RNA levels.

Next, integration of RNA and protein measurements for joint analysis (Methods) highlighted gene programs that are either common across conditions or unique to different conditions (Fig. 3i), with the co-cultured cells uniquely enriched for induction of immune escape pathway genes and the ICR signature. Specifically, we learned programs in each treatment condition by an adapted jackstraw principal component analysis (PCA) procedure (Methods; Extended Data Fig. 5a–g), and annotated them by enrichment for functional gene categories (Supplementary Table 5). Several programs, including cell cycle regulation, DNA repair and antigen presentation, were shared across conditions, whereas immune escape programs were uniquely recovered in co-culture data and interferon response genes in IFN-γ stimulation (Fig. 3i, Extended Data Fig. 5h and Supplementary Table 5). Thus, scRNA and protein profiles provide rich relevant phenotypes for assessing the impact of CRISPR perturbations.

A computational framework for analysis of Perturb-CITE-seq. We developed a computational framework to model the effects of genetic perturbations on both RNA and protein profiles of individual genes (Fig. 4a; Methods). Briefly, we used dial-out PCR data to determine the identity of perturbations (sgRNAs) in each cell (Extended Data Fig. 6a; Methods) and applied a linear model with elastic net regularization to infer the mean effect of each perturbation (sgRNA) on each feature (RNA and protein levels). We used a total of 4,461 RNA and 20 protein features, including all measured proteins and the union of the 1,000 top variable genes (Methods), and gene members of the programs identified by jackstraw PCA in any one condition. As we previously showed[1], detection of a sgRNA in a cell does not necessarily mean that this cell is perturbed, either because the sgRNA has not perturbed the gene or the perturbation does not have an impact. To address this, we calculated the probability of a successful perturbation after fitting an initial regulatory matrix, and then updated sgRNA assignments using an estimate of the probability for each cell that it was successfully perturbed. Our model extends our multiple-input, multiple-output, single-cell analysis (MIMOSCA) framework[1] by grouping sgRNAs targeting the same gene according to their concordant effects across features (Methods). Following our previous studies[1], we included both cell cycle and cell complexity (number of UMIs) as known covariates that impact cell profiles (Fig. 3f and Extended Data Fig. 4c). Inclusion of these covariates improved the model fit quality (residuals approach mean zero, independent and identically distributed; Extended Data Fig. 6b,c). Finally, we performed a permutation test to assess the empirical significance of each coefficient in the inferred regulatory matrix (Methods).

Perturb-CITE-seq identifies immune evasion programs/modules. The Perturb-CITE-seq model correctly reconstructed the impact of genes known to affect resistance to immunotherapy, in particular the effect of perturbing the IFN-γ response machinery (Fig. 4b), along with new pathways. The regulatory model can be parsed into eight major cofunctional modules of perturbations that similarly impact one or more of four major gene-coregulated



Condition	Program no.	Representative pathway	Key features
Control	3	DNA repair	*FEN1, MCM7, MCM10, BRCA1, PCNA*
IFN-γ	8	IFN-γ signaling (cytokine-mediated)	*IFITM3, SP100, STAT1, HLA-B, ADAR, ISG15, IFI35, SAMHD1, PSMB8, HLA-E, BST2, ISG20, IRF1, GBP2, XAF1, CD274* protein
Co-culture	0	Antigen processing via MCH-I pathway	*PSMD/PSMA/PSMB* family genes, *B2M, HLA-A, HLA-C, CD44* protein
Co-culture	3	Antigen processing and presentation via MHC-II pathway	*CENPE, HLA-DMA, RACGAP1, KIF4, HLA-DPB1, HLA-DRA, HLA-DRB1, HLA-DPA1*
Co-culture	5	Response to cytokine stimulus	*ITGB1, CDKN1A, CXCL8, CSF1, MT2A, SOX9, CCL2, IFNGR1, MMP1, IL1B, CD49f* protein
Co-culture	6	p53 DNA damage response	*CCNB1, PCNA, GADD45A, CDK1, E2F4, GTSE1, CD274* protein
Co-culture	9	Chemotaxis and migration in response to cytokines	*IL6, CXCL* genes, *ICAM1, CCL2*

Fig. 3 | Single-cell protein and RNA profiles reveal regulation of genes and programs involved in immune evasion. a,**b**, Distinct protein profiles across immune pressures highlight the regulation of cell surface proteins whose genetic perturbation confers resistance to TIL-mediated killing. **a**, Uniform manifold approximation and projection (UMAP) embedding of single-cell CITE-seq antibody count profiles (dots) colored by condition. **b**, $\log_2$(FC) (dot color) and significance ($-\log_{10}(P)$, dot size (and statistically significantly up-/downregulated encircled with black border) logistic regression model; Methods) between each pair of conditions (rows) of each cell surface protein (columns) measured by CITE-seq. **c**,**d**, Regulation of *CD58* (**c**) and *CD274* (**d**; PD-L1) by culture condition. Distribution of protein counts (*y* axis, left) or RNA (normalized expression; *y* axis, right; Methods) for *CD58* and *CD274*. $^{*}P<1\times10^{-10}$, $^{**}P<1\times10^{-50}$, $^{***}P<1\times10^{-100}$, $^{****}P<1\times10^{-150}$, Welch's *t*-test. Middle dot, median; box edges, 25th and 75th percentiles; whiskers, most extreme points not exceeding $\pm1.5\times$ interquartile range. Shading denotes a kernel density estimate with a bandwidth of 0.4. **e**–**g**, Variation in RNA profiles across and within conditions captures cell cycle state and MHC-I protein expression. UMAP embedding of scRNA-seq profiles (dots) colored by condition (**e**), cell cycle phase signature (**f**) or MHC (HLA-A,B,C) expression level (**g**, color bar). **h**,**i**, RNA expression of key immune genes and programs is impacted by increased immune pressure. **h**, $\log_2$(FC) (dot color) and significance ($-\log_{10}(P)$, dot size (and statistically significantly up-/downregulated encircled with black border))[31] between each pair of conditions (rows) of the RNA of selected immune genes (columns) measured by scRNA-seq, and differentially expressed between conditions. **i**, Gene programs identified by jackstraw PCA in each condition, representative enriched Gene Ontology processes and selected member genes. NS, not significant.

programs (Fig. 4c,d). Importantly, genes that were hits in the viability screen partitioned into different functional modules, thus highlighting the effect of the same (converging) or distinct (divergent) mechanism that could not be distinguished by viability screen alone. Those genes are also often members of regulated programs (Fig. 4d).

First, the model accurately recovered the effects of perturbing components of the IFN-γ response pathway. Perturbation of any major known node of this pathway downregulated a coherent regulatory program in co-culture (Fig. 4b). The downregulated program included key components of antigen presentation and associated machinery (for example, *PSMB4*, *PSMB8*, *PSMB9*, *PSMA4*, *HLA-A,B,C* gene and protein, *HLA-E*, *HLA-F* and *HLAD-DPB1*), chemokines associated with antitumor immunity (*CXCL1*, *CXCL2*, *CXCL8*, *CXCL10* and *CXCL11*), inflammatory cytokines (*STAT3*, *IL1B* and *IL6*), interferon response elements (*STAT1*, *IRF1*, *IRF3*, *IFITM3* and *IFIT6*), surface checkpoints (CD274 protein, *CD47* gene and protein) and genes associated with cell differentiation states (*SOX4*, *ITGA3* and *ITGA1*). Thus, multiple transcripts and proteins implicated in modification of the response to immunotherapies are directly regulated by the IFN-γ–JAK/STAT axis, suggesting that some of these mechanisms are a consequence of defective IFN-γ–JAK/STAT signaling rather than of independent modes of resistance.

Other transcripts and proteins were upregulated following perturbations in the IFN-γ–JAK/STAT nodes. These include *SERPINE2* and *TGFB2*, CD9 protein, CD59 protein and both *CD58* transcript and CD58 protein (Fig. 4b). It is likely that the induction of these genes is not through the IFN-γ–JAK/STAT pathway. For example, knockout or downregulation of *CD58* is associated with immune evasion (Figs. 2d,e and 3d) and thus its upregulation following perturbations in the IFN-γ–JAK/STAT module is probably not part of this pathway's immune evasion mechanism, and the role of *CD58* in immune evasion may therefore be distinct from defects in the IFN-γ–JAK/STAT pathway.

To focus on other distinct cofunctional modules, we examined the regulation matrix after exclusion of perturbations in IFN-γ–JAK/STAT pathway genes (Fig. 4c) and recovered new regulators and mechanisms of immune evasion either related to or distinct from the impact of the IFN-γ–JAK/STAT pathway (Fig. 4c,d, Supplementary Table 6 and Supplementary Note).

The perturbations also altered the expression of the ICR program that we originally defined in patients[1], in both co-culture (Fig. 4e) and in response to IFN-γ treatment (Fig. 4f). Perturbations of the IFN-γ–JAK/STAT module strongly increased the overall expression of the ICR signature (*JAK1*, $P = 1.3 \times 10^{-10}$; *IFNGR2*, $P = 1.4 \times 10^{-8}$; *IFNGR1*, $P = 3.3 \times 10^{-5}$; *STAT1*, $P = 9.6 \times 10^{-5}$; statistics derived by Welch's *t*-test, Methods), as did deletion of *TMEM173*, encoding STING (STimulator of INterferon Genes), which activates a type I interferon response[25] (Fig. 4e). STING agonists are currently under evaluation in clinical trials in patients with melanoma who have resistance to ICI therapy, and in other cancers[26]. In contrast, other perturbations, including KO of *CDK6*, *MYC*, *ILF2*, *DNMT1* or *ACSL3*, repressed the ICR signature (Fig. 4f), consistent with our previously reported patient and preclinical data where we demonstrated that upregulation of *MYC* and *CDK4/6* (and their transcriptional targets) was associated with increased resistance to immunotherapy, and that CDK4/6 inhibitors reduced resistance in human and preclinical melanoma models[9].

Taken together, our Perturb-CITE-seq analysis rediscovered key mechanisms of immune evasion, organized them into modules and related them to the genes and programs they impact; it also recovered new modules that may confer immune evasion beyond defects in the IFN-γ–JAK/STAT pathway and antigen presentation.

Loss/downregulation of *CD58* confers cancer immune evasion. From our analyses across both viability and Perturb-CITE-seq data, *CD58* emerged as a compelling mechanism of immune evasion: KO of *CD58* conferred resistance to T-cell killing in the viability screen, and CD58 RNA/protein were downregulated in cells surviving TIL co-culture. Importantly, *CD58* is not activated by the IFN-γ–JAK/STAT pathway (Fig. 4b) and *CD58* KO conferred a different molecular phenotype than IFN-γ–JAK/STAT pathway perturbations, belonged to a distinct module and did not impact the expression of antigen-presentation genes (Fig. 5a). Thus, *CD58* loss may represent a resistance mechanism distinct from IFN-γ–JAK/STAT pathway inactivation. CD58 is an adhesion protein typically expressed on the surface of antigen-presenting cells, where it binds to CD2 on $CD8^+$ T cells and NK cells[22]. Little is known about the potential role of *CD58* in cancer, partly because there is no known mouse homolog in which to study it in preclinical models. Thus, patient-derived melanoma/TIL co-cultures provide a unique opportunity to study *CD58* in the context of human immune evasion.

To validate the role of *CD58* in immune evasion, we generated individual KO of *CD58*, *B2M* or *PD-L1* (Methods) in three patient-derived melanoma cell lines and performed co-cultures with their autologous TILs (Extended Data Fig. 7a–c; Methods). In all models, *CD58* KO and *B2M* KO conferred resistance (Fig. 5b–d) while *PD-L1* KO sensitized cells to T-cell-mediated killing (Fig. 5b). Importantly, we excluded the possibility that these differences were due to differences in either proliferation rates (Extended Data Fig. 7d) or the apoptotic potential of different KO models (Extended Data Fig. 7e). Next, we performed competition assays where blue fluorescent protein (BFP)-expressing parental and red fluorescent protein (RFP)-expressing KO cell lines were mixed in a 1:1 ratio and co-cultured with TILs in the same well (Fig. 5e). Both *CD58* KO and *B2M* KO cells had a strong competitive advantage over *CD274* KO and parental cell lines (Fig. 5f). Loss of *CD58* also conferred resistance to NK-cell-mediated killing (Fig. 5g, Extended Data Fig. 7f and Supplementary Note).

Loss of *CD58* is an orthogonal mechanism of immune evasion. Because *CD58* KO conferred immune evasion of both T and NK cells, we hypothesized that its mechanism of action is independent of antigen presentation via MHC. Notably, in our Perturb-CITE-seq data, *CD58* KO did not substantially alter the level of the *B2M* transcript or MHC Class I protein (encoded by *HLA-A*) (Fig. 5h). Using flow cytometry we show that, compared to parental cells, *CD58* KO did not alter expression of MHC proteins at baseline or their induction in response to IFN-γ (Fig. 5i,j). Our Perturb-CITE-seq data also suggested that *CD58* KO led to increased expression of PD-L1 (Fig. 5h). Using flow cytometry, we confirmed that stimulation with low- or high-dose IFN-γ resulted in higher levels of PD-L1 protein in *CD58* KO cell lines compared to parental control (Fig. 5k), suggesting that upregulation of PD-L1 could contribute to immune evasion in *CD58* KO in T-cell co-culture.

Conversely, neither *B2M* nor *HLA-A* KO impacted CD58 protein levels in our Perturb-CITE-seq screen (Fig. 5h and Extended Data Fig. 7a). Indeed, defects in the IFN-γ–JAK/STAT nodes led to increased *CD58* RNA and protein expression (Figs. 4b and 5h) and stimulation with IFN-γ (at either 1 or 10 $ng\,ml^{-1}$) did not increase protein abundance of CD58 (Extended Data Fig. 7g–j and Fig. 3b), suggesting that *CD58* is not induced via the IFN-γ pathway.

Finally, we determined the level of mRNA expression of *CD58* in patients with melanoma with resistance to ICI. Compared to malignant cells from treatment-naive patients, ICI-resistant patients had a significantly lower expression of *CD58* (Fig. 5l).

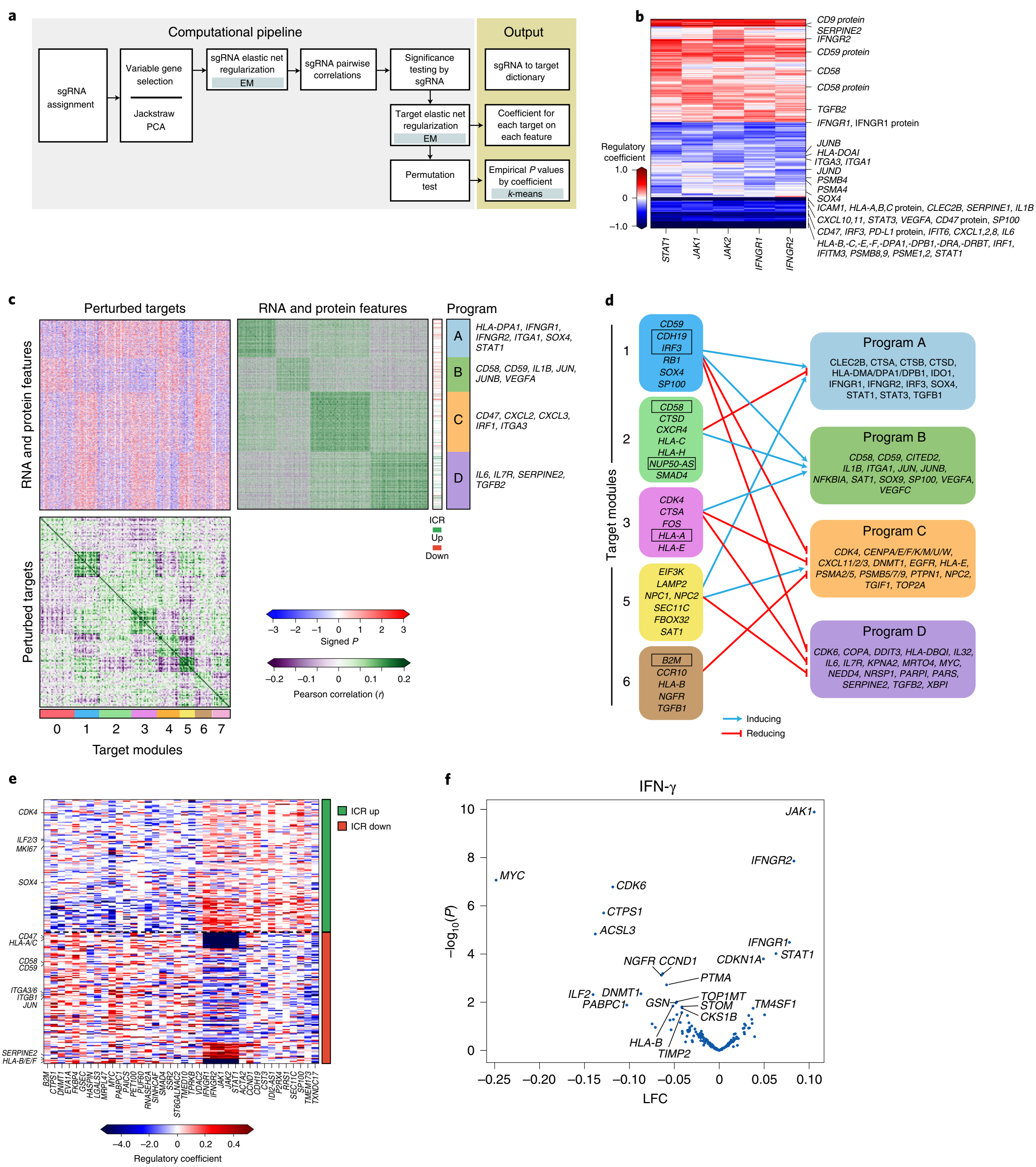
Together, these data suggest that downregulation or loss of *CD58* represents a clinically relevant immune evasion mechanism that is

Fig. 4 | Perturb-CITE-seq reveals cofunctional modules that are dependent or independent of a predominant IFN-γ–JAK/STAT mechanism. a, Overview of computational approach (Methods). **b**, Perturbations in the JAK/STAT pathway affect both known and putative mechanisms of immune evasion. Regulatory effect (β values) on RNA and protein features (rows) following perturbation of different genes in the JAK/STAT pathway (columns). **c**,**d**, Cofunctional modules and coregulated programs in the Perturb-CITE-seq screen. **c**, Middle heatmap: signed significance ($-\log_{10}$(empirical P) × sign(β), red/blue color bar) for the effect on each RNA/protein feature (rows) of perturbation of each gene (columns, excluding JAK/STAT targets) in the co-culture condition. Right and bottom matrices: Pearson correlation coefficient (purple/green color bar) between the significance profiles of either gene/protein features (right matrix) or perturbed genes (bottom matrix). Cofunctional modules (bottom bar) and coregulated programs (right bar) are identified by separate k-means clustering of the bottom and right matrices ($k = 4$ and 8, respectively), and clustering defines the row and column order. **d**, Representation of regulatory connections between selected modules (left) and programs (right) from **c**. Bold font denotes selected regulators that are also members of programs; boxed text denotes selected regulators significantly enriched/depleted in the viability screen (Fig. 2d,e). Notably, the edges in **d** are opposite the sign of the empirical P value in **c**: a target module connected to a coregulated program with negative signed P values in **c** activates the corresponding program in **d**. **e**,**f**, The ICR program is coherently regulated by the perturbed regulators. **e**, Regulatory effect (β values) on RNA/protein features from the ICR program (rows, ICR as defined in ref. [9]) by perturbations of different genes in the screen (columns), clustered by k-means ($k = 2$). **f**, Change in ICR signature scores and their associated significance (y axis, $-\log_{10}(P)$, Welch's t-test) for each perturbation (dot) in the IFN-γ condition (Methods). Key perturbations with significant effects are noted.

orthogonal to those previously described involving defects in antigen presentation and IFN-γ response.

Discussion

In this study we developed Perturb-CITE-seq, a pooled CRISPR–Cas9 screen with multimodal, single-cell profiling readout and used it in patient-derived, tumor-immune models to dissect cancer-cell-mediated modulators of T-cell-mediated killing. A proof-of-concept study developed ECCITE-seq to show simultaneous detection of surface proteins, RNA and gRNAs in single cells[5], and our work presents a parallel methodology, a large genetic screen utilizing multimodal, single-cell protein and RNA detection and a computational framework for integrated multimodal perturbation analysis. We used a broadly applicable computational and statistical framework for integrated analyses of such screens, which accounts for key covariates including cell quality and cell cycle status, focuses



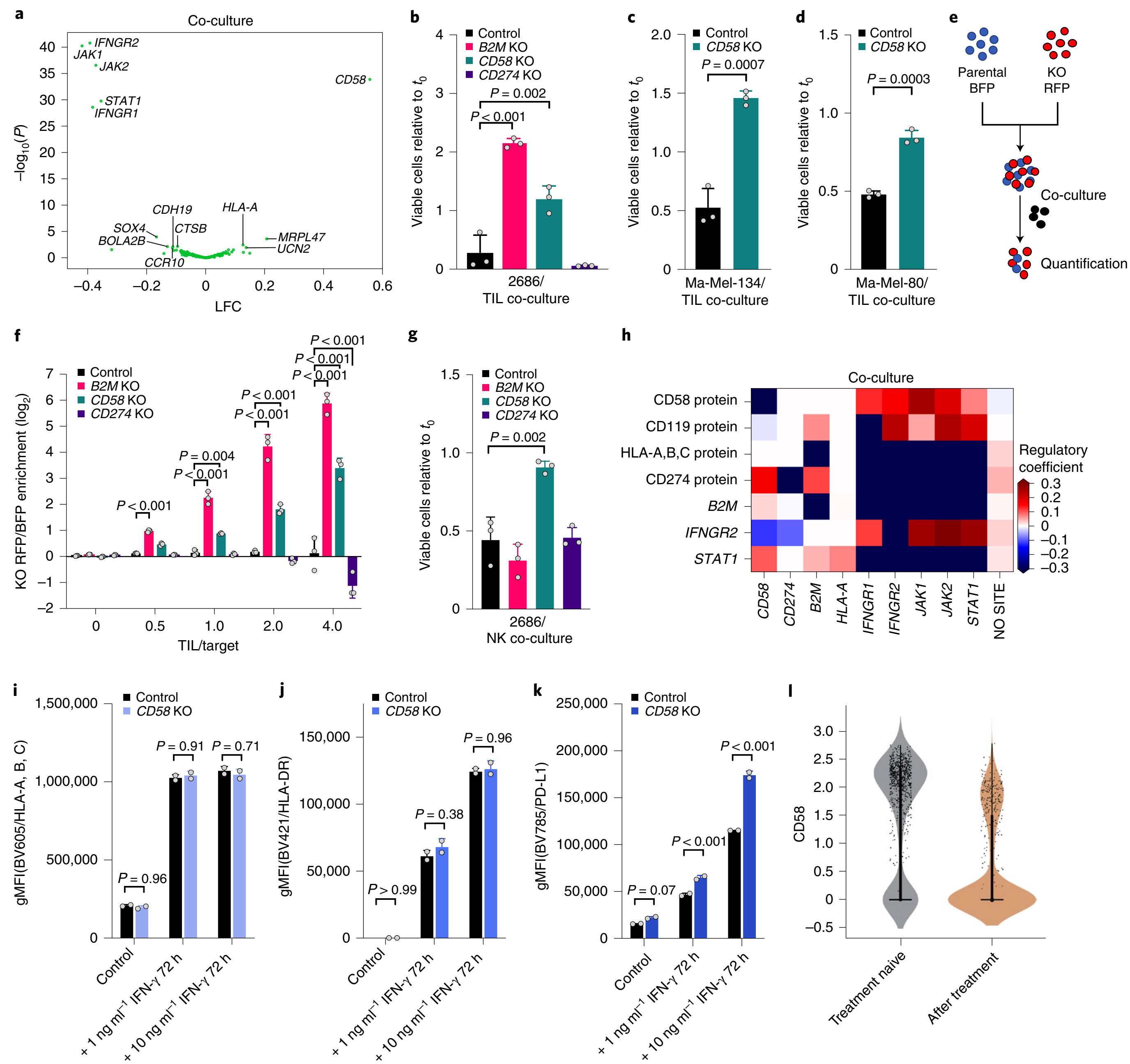


Fig. 5 | CD58 loss is a distinct mechanism of immune evasion from TIL- and NK-cell mediated killing. a, *CD58* perturbation affects a distinct regulatory program. Change in signature scores of the CD58 regulatory program (*x* axis, LFC) and its associated significance (*y* axis, $-\log_{10}(P)$, Welch's *t*-test) for each perturbation in the co-culture. **b–d**, Ratio of viable cancer cell lines 2686 (**b**), Ma-Mel-134 (**c**) and Ma-Mel-80 (**d**) (*y* axis, relative to t_0) in co-culture models of control, *CD58* KO, *B2M* KO and *CD274* KO cells. **e**, Competition assay schematic. BFP-labeled parental cells and RFP-labeled KO cells were co-cultured with TILs; the RFP/BFP ratio is calculated as an estimate of relative fitness. **f**, Competition assay of parental cells and matched *B2M* KO, *CD58* KO or *CD274* KO after 48 h of co-culture. **g**, Ratio of viable cancer cells in a NK co-culture model of parental cells compared to matched indicated genotypes. **h–k**, *CD58* perturbation in co-culture does not affect B2M and HLA expression at the RNA and protein level but induces *CD274*. **h**, Regulatory effect (β values from the model shown in **a**; red/blue: perturbation induces/represses gene features) in Perturb-CITE-seq on key RNA and protein (CITE-seq) features (rows) following perturbation of different genes in the JAK/STAT pathway, *CD58* or *CD274* (columns). **i–k**, Surface expression (gMFI, geometric mean fluorescence intensity) of MHC Class I (**i**), MHC Class II (**j**) or *CD274* (**k**) at baseline and following stimulation with different levels of IFN-γ for 72 h, in parental and *CD58* KO cells. **l**, Distribution of expression levels (*y* axis, $\log_2$(transcripts per million (TPM) + 1)) of *CD58* RNA in melanoma cells from tumors in patients who were either treatment naive (gray) or were resected after failure of immunotherapy (tan) in scRNA-seq data from the ICR signature discovery cohort[9]. **b**,**f**,**g**, One-way ANOVA with Tukey's post hoc test; **c**,**d**, two-sided *t*-test; **i–k**, error bars, mean ± s.d. All experiments were performed in triplicate (except **i–k**, in duplicate) in each of at least two independent experiments.

on cells harboring impactful perturbations, generates a detailed regulatory model and partitions it to interpretable cofunctional modules that control coregulated programs.

Our rich, multimodal screen is critical to dissecting which genes are part of a shared mechanism of resistance, which represent distinct mechanisms and through which gene programs each act. First, Perturb-CITE-seq data readily identified major known clinical mechanisms of immune evasion, especially perturbations and their associated cell programs in the IFN-γ pathway and downregulation or defects of the antigen-presentation axis. Perturbations of different nodes within the IFN-γ pathway led to highly converging molecular phenotypes, irrespective of the level of the defect. This is consistent with genomic profiling in patients with resistance to therapy with either anti-CTLA-4 (ref. [27]) or anti-PD-1[7], and further emphasizes the role of IFN-γ in T-cell-mediated antitumor immunity. Among genes downregulated by IFN-γ pathway perturbations, including *CXCL10* and *CXCL11*, are those whose downregulation was previously associated with immune evasion, suggesting that these do not represent a salient mechanism of immune evasion. In contrast, other genes whose perturbation enhanced immune evasion in our screen (for example, *CD58* and *CD59*) appear to reflect distinct mechanisms, because their expression was not impaired by IFN-γ signaling defects and because the molecular phenotype following their perturbation is distinct. Because many gross phenotypes in biology (immune evasion, cell cycle, viability, differentiation and so on) are affected by multiple pathways and involve interactions between cells, this approach should provide a powerful solution in other systems.

In particular, our Perturb-CITE-seq and viability screens highlighted *CD58* as a gene whose knockout enhances resistance to T-cell-mediated killing, as a member of a cofunctional module with a distinct phenotype and as a target whose RNA and protein levels are reduced in co-culture but not activated by the IFN-γ pathway. Because there is no known mouse homolog of *CD58*, this target was not discovered in previously reported CRISPR screens performed in mouse models[11–13]. While it was a top-ranking hit in an immunotherapy screen in a human-engineered system[10], its role remained poorly understood. In patients with diffuse large B-cell lymphoma[28], *CD58* mutations were concurrent with those in *B2M* leading to loss of antigen presentation. Since the two mutations frequently occurred concurrently, it remained unclear whether these are independent mechanisms of immune escape. Our experiments and analysis show that loss of *CD58* confers immune evasion to a similar extent as loss of MHC Class I expression itself (through *B2M* KO), but via an independent path and without impacting the expression of antigen-presentation genes and proteins (Extended Data Fig. 8). Furthermore, PD-L1 is upregulated in *CD58* KO, suggesting that *CD58* loss could confer immune evasion either directly (reduced adhesion) or indirectly (inhibitory PD-L1) (Extended Data Fig. 8). *CD58* KO also impacted NK-mediated killing (Fig. 5g), which may have implications for NK-cell-based therapies in tumors with loss of antigen presentation. Notably, the CD58–CD2 axis is a potent costimulatory pathway in $CD8^+$ T cells that lack expression of CD28 ($CD8^+CD28^-$ T cells)[29], which are common in the tumor microenvironment and peripheral blood of patients with solid tumors[30], and were the predominant T cells in our Perturb-CITE-seq screen (Extended Data Fig. 4b). We speculate that *CD58* downregulation or loss may contribute to immune evasion through distinct mechanisms, including loss of T-cell co-stimulation, increased expression of co-inhibitory PD-L1 and possibly reduced T-cell infiltration due to impaired T-cell adhesion. Due to the complex regulation of PD-L1 protein stability and the poorly understood kinetics of CD58 protein, additional studies evaluating CD58–PD-L1 balance are necessary to dissect their functional relationship.

Our study presents a large-scale CRISPR–Cas9 screen with multimodal, single-cell readouts providing a general approach, and addresses a critical clinical challenge in a unique patient-derived cell culture model system. We recover the landscape of resistance mechanisms to immunotherapies and, guided by our high-dimensional data, validate a new mechanism of immune resistance. Large-scale, multimodal screens spanning RNA, proteins, chromatin accessibility and imaging features should enable unprecedented discovery across diverse biological systems, and detailed dissection of complex cellular and intercellular circuits.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41588-021-00779-1.

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Methods

Patient-derived melanoma cell culture. Patient-derived melanoma cell lines were grown on nonpyrogenic, polystyrene tissue-culture-treated plastic ware (Corning) in RPMI 1640 medium supplemented with 10% heat-inactivated fetal bovine serum, GlutaMax, 10 mM HEPES, 10 mg l^{-1} insulin, 5.5 mg l^{-1} transferrin, 6.7 µg l^{-1} sodium selenite and 55 µM 2-mercaptoethanol (all Thermo Fisher Scientific) at 37 °C and 5% CO_2 in a humidified incubator. Melanoma cell line 2686 and matched TILs were previously derived (under institutional review board protocol no. 2004-0069) and provided by MDACC[9,14]. Melanoma cell lines Ma-Mel-80d (a subpopulation of Ma-Mel-80) and Ma-Mel-134 and corresponding TILs were provided by UK-Essen. Melanoma cells tested repeatedly negative for contamination with mycoplasma and other contaminants using PlasmoTest (InvivoGen).

Miniaturized imaging-based co-culture assay. Melanoma target cells expressing nuclear localization signal (NLS)-dsRed were seeded (10^4 per well) in a black-walled, 96-well plate (Corning) and treated for 16 h with 1 ng ml^{-1} recombinant human Interferon gamma (Peprotech). After 16 h, medium was replaced with 100 µl of full melanoma medium, without phenol red, containing 4 µM Caspase-3/7 activity dye (CellEvent, Thermo Fisher Scientific). Plates where imaged using a Celigo Imaging Cytometer (Nexcelom) to obtain time point 0 (t_0) cell counts. TILs were thawed for 3 d before co-culture and either cultured in TIL medium with 3,000 IU ml^{-1} of IL2 or reactivated on plates coated with 100 ng ml^{-1} OKT3 in PBS when comparing preactivated to rested TILs. On the day of assay, TILs were collected, centrifuged and resuspended in TIL medium without IL2, counted and increasing ratios of TILs were added to melanoma target cells to a total volume of 200 µl per well (final caspase activity dye, 2 µM); co-cultures were incubated at 37 °C and 5% CO_2 in a humidified incubator. The plates were reimaged 24, 48 and 72 h after TIL addition. Viable target cells were counted using Celigo Imaging Cytometer software. First, cells were identified by the size and intensity of the NLS-fluorescent protein. Debris was defined by adjusting gating parameters to exclude condensed, late apoptotic cells and Caspase-3/7 dye intensity was measured across all cells in the green channel to identify apoptotic cells. Finally, viable cells were defined as 'viable Cells (Class) = NLS-dsRed positive AND NOT debris size AND NOT Caspase-3/7 positive'. Viable target cell counts were normalized to respective well counts at t_0 using Excel (Microsoft) and plotted using GraphPad Prism 8 (GraphPad). To investigate the MHC Class I dependency of TIL-mediated lysis of melanoma target cells, we performed co-culture experiments in the presence of MHC Class I blocking antibody (final concentration, 50 µg ml^{-1}, clone W6/32; Thermo Fisher Scientific) added 2 h before addition of TILs. To determine the ratio at which MHC Class I blocking antibodies were tested, we tested several TIL/tumor ratios and performed blocking experiments at the ratio with optimal MHC/TCR-dependent tumor lysis over potential bystander effects. For mixing studies of single-gene KO with wild-type cells, KO were generated in NLS-dsRed target cells mixed with unmodified control target cells expressing NLS-BFP. Viable cells were identified as outlined above, and fold enrichment was calculated as dsRed (viable)/BFP (viable) and normalized to dsRed (viable)/BFP (viable) in conditions without TILs.

Large-scale co-culture assay for CRISPR–Cas9 viability screen. Melanoma cells were transduced with an ICR library extended with an equal number of control sgRNAs to increase power for enrichment detection, as described in 'Generation of pooled Perturb-CITE-seq gDNA library' (Supplementary Note), and three pellets of 1×10^6 perturbed target cells were collected on day 7 for subsequent quantification of dropout of essential genes. After 14 d, perturbed melanoma cells were plated in 12-well plates at 1.25×10^5 cells per well (~900× guide representation per plate). Simultaneously, three aliquots of 1×10^6 melanoma cells were collected to quantify the distribution of sgDNAs at the beginning of the experiment. Cells were treated with 1 ng ml^{-1} recombinant human IFN-γ (in both treatment and co-culture conditions) or left untreated and incubated at 37 °C and 5% CO_2 in a humidified incubator. After 16 h, the medium was replaced with fresh prewarmed medium without IFN-γ. TILs had been thawed 3 d before the experiment and cultured in TIL medium supplemented with 3,000 IU ml^{-1} rhIL2. On the day of TIL addition, TILs were resuspended, pelleted at 400*g* at 4 °C for 5 min, resuspended in TIL medium without IL2, counted and added at an effector/target ratio ranging from 1:1 to 4:1 to the perturbed IFN-γ-pretreated melanoma target cells in a final volume of 2 ml. Plates were centrifuged at 400*g* for 5 min to ensure cell-to-cell contact, and co-cultures were kept for 48 h at 37 °C and 5% CO_2 in a humidified incubator. For the untreated control and IFN-γ treatment, the experiment was set up in triplicate plates. For each of the co-culture ratios one plate was set up. After 48 h, each plate was washed once with ice-cold PBS to remove TILs and dead cells; surviving cells were detached using Accutase, pelleted and pellets were stored at −80 °C until further processing. Surviving target cells on distinct plates containing all conditions as triplicate were counted to quantify target cell killing and verify accurate immune selection pressure, as previously determined in the miniaturized co-culture experiments. Genomic DNA was extracted and purified from cells using the QIAamp DNA Micro Kit (Qiagen, no. 56304) with the user-developed protocol QA43 and no more than ~0.5 M cells per column. sgDNA sequences were amplified from genomic DNA for Illumina sequencing in two PCR steps. The first was conducted in parallel 50-µl reactions, ensuring no more than 500 ng of genomic DNA template per reaction. Using primers 642 F and 643 R and NEBNext High-Fidelity 2× PCR Master Mix (NEB M0541L), PCR was run for ten cycles with the following parameters: 98 °C denaturation for 60 s, 68 °C annealing for 30 s and 72 °C extension for 60 s. Taking 5 µl of PCR1 for another 50-µl reaction, using primers 997 and 998 to add Illumina adapters and NEBNext master mix, PCR2 was run for ten cycles with the following parameters: 98 °C denaturation for 15 s, 62 °C annealing for 15 s and 72 °C extension for 16 s. With 5 µl of PCR2 product, quantitative PCR was conducted with SYBR green and the same primers and parameters used for PCR2, to estimate the number of PCR cycles required to reach sufficient amplification for Illumina sequencing. With the same primers and parameters as PCR2, additional cycles were run for each reaction as calculated. PCR products were then purified by 1× SPRI and prepared for sequencing on an Illumina HiSeq 2500 in RapidRun mode with custom read 1 (primer 503 F). All primer sequences are listed in Supplementary Table 8.

Viability screen analysis. CRISPR–Cas9 screens from genomic DNA were analyzed using software packages MAGeCK and MAGeCKFlute[32]. Briefly, MAGeCK maps sequencing reads to the reference library of sgDNA and returns the quantity of reads that confidently mapped to each sgDNA sequence with no allowance for mismatches. To control for sequencing depth, sgDNA counts were median normalized. Several metrics were used to evaluate the quality of each sequenced sample, including Gini index, missed gDNAs and correlation of sgDNA counts between replicates. MAGeCK's robust rank aggregation method was used to discover significantly enriched or depleted genes in test versus control conditions.

Perturb-CITE-seq co-culture experiment. After 14 d of editing and selection, 1.25×10^5 Cas9-expressing, ICR library, perturbed melanoma target cells were seeded per well of a 12-well plate (one plate with a total of 1.5×10^6 cells per condition, >1,800× representation). Cells were either treated with 1 ng ml^{-1} recombinant human IFN-γ (in both treatment and co-culture conditions) or left untreated and incubated at 37 °C and 5% CO_2 in a humidified incubator. After 16 h, the medium was replaced with fresh prewarmed medium for the co-culture group and cells of both control and treatment groups were washed once with cold PBS, detached using Accutase (Stem Cell Technologies), quenched with fully cold medium, pelleted at 400*g* at 4 °C for 5 min, resuspended in CITE staining buffer (PBS plus 2% BSA and 0.01% Tween-20), filtered through a prewetted 70-µM cell strainer (Corning) and counted; 1×10^6 cells were then aliquoted in 1.5-ml DNA LoBind microcentrifuge tubes (Eppendorf, Hamburg, Germany) and pelleted again. Cells were resuspended in 100 µl of CITE staining buffer and 5 µl of TruStain FcX blocking antibodies (Biolegend) was added, followed by incubation for 10 min on ice. Cells were again pelleted and resuspended in 100 µl of CITE staining buffer and 5 µl of CITE-seq antibody pool (~1:500 final dilution for each antibody) was added to each sample, followed by incubation for 30 min on ice. Next, cells were washed for a total of three washes with 100 µl of CITE staining buffer and then processed for scRNA-seq, with 15,000 cells loaded onto each of eight channels per condition using the 10X Chromium system with the Chromium Single Cell 3' Library and Gel Bead kit v.3 (10X Genomics), as per the manufacturer's instructions.

For the co-culture condition, TILs had been thawed 3 d before the experiment and cultured in TIL medium supplemented with 3,000 IU ml^{-1} rhIL2. On the day of TIL addition, TILs were resuspended, pelleted at 400*g* at 4 °C for 5 min, resuspended in TIL medium without IL2, counted and 2.5×10^5 cells were added in a final volume of 2 ml to perturbed IFN-γ-pretreated melanoma target cells (final effector/target ratio, 2:1). Plates were centrifuged at 400*g* for 5 min to ensure cell-to-cell contact and co-cultures were maintained for 48 h at 37 °C and 5% CO_2 in a humidified incubator. After 48 h, medium was removed from co-culture plates and the cells were washed once with cold PBS and detached as described above. After initial centrifugation, cells were resuspended in ice-cold PBS with a 1:500 dilution of Zombi-NIR dead cell stain (Biolegend) and incubated for 15 min in the dark on ice. Cells were then washed once with CITE staining buffer, resuspended, filtered, counted, aliquoted and flow cytometry blocked as described in Loss of CD58 is an orthogonal mechanism of immune evasion. CD45-Pacific Blue (HI30, Biolegend, no. 304022) was included during CITE antibody staining to label TILs for sorting. After three washes in CITE buffer, cells were again filtered and viable melanoma target cells sorted using a FACS Aria III with a cooling system. Melanoma target cells were identified using cell gating with FSC-A and SSC-A, doublet exclusion using FSC-A and FSC-H, gating on Zombi-NIR dim cells and then gating on $CD45^-$, $mKATE2^+$ target cells. After sorting, cells were centrifuged once at 400*g* for 5 min at 4 °C, resuspended in CITE staining buffer and 15,000 cells per channel were loaded onto the 10X Chromium system as described above.

Normalization and integration of single-cell data. Data on scRNA-seq counts waere normalized to 1,000,000 total counts per cell (transcripts per million), followed by a natural log transformation. CITE-seq data were normalized according to the formula[33]:

$$\max\left(\ln\left(\frac{c_a+1}{c_c+1}\right),0\right)$$

where c_a is UMI count for antibody a and c_c is UMI count for its corresponding IgG control c (Supplementary Table 3). The two modalities were integrated by concatenation of their normalized count matrices for further analysis.

Computational model for Perturb-CITE-seq analysis. To infer a model of gene regulation from Perturb-CITE-seq data, following initial sgRNA assignment to cells we performed feature selection. For each of the three conditions (control, co-culture and IFN-γ), 1,000 highly variable genes were selected using the Scanpy v.1.4.4 implementation of highly variably gene selection[34]. In addition, all features from each condition's ten most significant Jackstraw PCA programs were included and all 20 CITE antibodies. The union was taken across all conditions, resulting in one set of 4,481 features used for each experimental condition.

Next, for each condition separately, we learned a linear model to predict the effect of each sgRNA on each feature using the Scikit-learn implementation[35] of elastic net regularization with the following parameters: l1_ratio = 0.5, alpha = 0.0005, max_iter = 10,000.

Fit quality was assessed by correlation of model residuals. Inclusion as covariates of cell quality (defined as the total number of UMIs in each cell) and cell cycle state (assigned with Scanpy's implementation of scoring cell cycle genes[34]) improved the fit. Following the first elastic net regularization, an expectation maximization (EM)-like procedure was run along the columns of the sgRNA assignment matrix to account for false-positive sgRNA assignments and sgRNAs that did not perturb their target, using our previously published MIMOSCA framework[1]. Briefly, each cell is modeled as being derived from either a perturbed or unperturbed population. For each sgRNA, the elastic net regularization is rerun with perturbed cells assigned to the unperturbed population. The model is run twice: once with cells assigned to the perturbed population and once with cells assigned to the unperturbed population. A sum of squares error (SSE) is calculated for each of these two models. The difference in SSE between these two models is transformed to a probabilistic estimate that the cell originated from the perturbed population via a logistic function. If the SSE is far greater in the model with the cell assigned to the perturbed population, sgRNA assignment is updated to be near zero. Conversely, if the SSE is larger in the model with the cell assigned to the unperturbed population, sgRNA assignment is maintained near 1. Running this procedure across all sgRNAs effectively transforms the binary sgRNA assignment matrix to a probabilistic estimate of successful perturbation. Following this procedure, elastic net is rerun with the probabilistic sgRNA assignment matrix. The procedure is iterated only once (and thus is not a full EM). A full EM approach is not guaranteed to converge to a global minimum in a high-dimensional linear regression problem with noisy and missing data[36]. Empirically, addition of a second iteration had only a very modest effect on sgRNA/target assignments. For example, in the co-culture condition, a single iteration of EM adjusted the assignment of 1,741 targets to near zero (confidence that the cell contains an effective sgRNA <25%), while an additional iteration adjusted only 97 target cell assignments.

The regulatory matrix fit by this procedure was used to assess concordant effects across sgRNAs with the same target, by calculating the pairwise correlation between the regulation profiles for sgRNAs across all features. Pearson correlation of sgRNAs with different targets was then used to calculate a synthetic null distribution of discordant perturbations, and pairwise Pearson correlation of sgRNAs with the same target was transformed into an empirical *P* value followed by Benjamini–Hochberg multiple hypothesis false discovery rate (FDR) correction. To aggregate from sgRNAs to the gene level, sgRNAs with correlated activity were mapped to their target gene to construct a sgRNA-to-target dictionary. A lenient FDR threshold of 0.50 was used due to the large feature space and the sparsity of elastic net regularization. This resulted in 141, 183 and 181 gene-level targets (among 248 total target genes) for the control, IFN-γ and co-culture conditions, respectively. All nontargeting sgRNAs were collapsed to the same target while intergenic sgRNAs were kept separate as individual negative controls. Note that IFN-γ–JAK/STAT perturbations have been removed from Fig. 4c for visualization purposes only. The computational pipeline was run in its entirety with all perturbations. Removal of these highly impactful perturbations allowed clustering features into programs based on their response to perturbations across our target library.

The sgRNA assignment matrix was mapped to a binary target assignment matrix using the sgRNA-to-target dictionary for the next iteration of elastic net regularization. An EM-like procedure was then rerun according to that shown above, except at the target rather than the sgRNA level. A final elastic net regularization was run following the reassignment step.

A permutation test was used to assess the empirical significance of regulatory coefficients as previously described[1]. Briefly, for each target, the assignment of cells to targets was randomly permuted followed by recomputing the linear model. This procedure was run separately for each target, each with 10,000 random permutations, to calculate target-wise null distributions of coefficient values. The coefficients of the model with correct cell assignment were then transformed to empirical *P* values. This procedure partially corrects for the bias of sample size and effect size in regulatory coefficients, and enables direct comparison of empirical *P* values across targets.

Empirical *P* value matrices were clustered using the Scikit-learn implementation of *k*-means clustering. The number of clusters was determined using the 'elbow method' based on the inertia of the fit sweeping the number of clusters from two to 20.

Single-gene KO in patient-derived cells using nucleofection of Cas9 ribonucleoproteins. Virus-free KO human melanoma cell lines were generated using nucleofection of Cas9 ribonucleoproteins (RNPs). Target sequences were derived from the original ICR library used to generate the CROP-seq library, and are listed individually for perturbed targets in Supplementary Table 8. To form RNPs, equimolar ratios of CRISPR RNA (IDT) were incubated with transactivating CRISPR RNA (IDT) at 95 °C for 5 min in nuclease-free duplex buffer, and thereafter cooled to room temperature to form gRNA complexes. Recombinant Cas9 enzyme (MacroLab) was mixed with gRNA at a 1:10 molar ratio and incubated at 37 °C for 15 min to form RNP complexes. Meanwhile, human melanoma cell lines were detached, washed once with PBS and 5×10^4 cells were resuspended in 20 µl of SF electroporation buffer prepared with SF supplement (Lonza). RNP complex solution (3 µl) was mixed with the cells and these were then nucleofected using program DJ-110 on a 4D-Nucleofector (Lonza) with 16-well strips. Cells were immediately recovered in full melanoma medium, seeded in 12-well plates and expanded.

Flow cytometry analysis of surface proteins after IFN-γ stimulation. Both 2686 control and *CD58* KO cells were stimulated with 1–10 ng ml^{-1} IFN-γ for 72 h and then detached with Accutase. Dead cells were labeled using Zombie-NIR (Biolegend) according to the manufacturer's instructions. Surface antigens were stained on ice using the following antibodies (all Biolegend): HLA-DR-BV421 (L243), HLA-A,B,C-BV605 (W6/32), CD274-BV785 (29E.2A3) and CD58-APC (TS2/9). Cells were fixed in fixation buffer (Biolegend) and analyzed on an Aurora Spectral Analyzer (Cytek Biosciences). The samples were than analyzed using FlowJo (FlowJo). For quantification of surface proteins. cells were gated based on FSC and SSC, single cells were selected using FSC-A and FSC-H and viable cells were selected using low Zombie-NIR fluorescence. To compare signal intensities of surface markers, geometric mean fluorescence intensity was used and all samples were run in duplicate.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

Processed data are available via the single-cell portal: https://singlecell.broadinstitute.org/ single_cell/study/SCP1064/ multi-modal-pooled-perturb-cite-seq-screens-in-patient- models-define-novel-mechanisms-of-cancer-immune-evasion. Raw data are available on the Broad Data Use and Oversight System through the accession code DUOS-000124: https://duos.broadinstitute.org.

Code availability

Code is available at https://github.com/klarman-cell-observatory/Perturb-CITE-Seq.

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Author contributions

B.I. and A. Regev conceived this project and provided overall supervision. J.C.M., P.I.T., K.R.G.-S., P.H., A.M.L., S.M., M.S.C., M.Z., C.R.A., M.R. and L.H. performed experiments. C.J.F., J.C.M., P.I.T., K.R.G.-S., B.C. and L.J.-A. performed analyses. A.

Rotem, K.W.W., B.E.J. and O.R.-R. contributed to project coordination. C.B. and D.S. provided critical patient models. C.F.J., J.C.M., P.I.T., K.R.G., A. Regev and B.I. wrote the manuscript. All authors reviewed and approved the final manuscript.

Competing interests

A. Regev is a cofounder and equity holder of Celsius Therapeutics, an equity holder in Immunitas and, until 31 July 2020, was an SAB member of Thermo Fisher Scientific, Syros Pharmaceuticals, Neogene Therapeutics and Asimov. From 1 August 2020, A. Regev has been an employee of Genentech. B.I. is a consultant for Merck and Volastra Therapeutics. B.E.J. is on the scientific advisory board of Checkpoint Therapeutics. From 16 November 2020, K.G.S. and O.R.-R. have been employees of Genentech. A. Rotem is a consultant to eGenesis, a SAB member of NucleAI and an equity holder in Celsius Therapeutics. Since 31 August 2020, A. Rotem has been an employee of AstraZeneca. K.W.W. serves on the scientific advisory board of TCR2 Therapeutics, T-Scan Therapeutics, SQZ Biotech and Nextechinvest and receives sponsored research funding from Novartis. K.W.W. is a cofounder of Immunitas, a biotech company. These activities are not related to the research reported in this publication. C.B. is on the scientific advisory board of Myst Therapeutics. A. Regev, B.I., J.C.M., K.R.G.-S., P.I.T. and C.J.F. are inventors on a patent application related to this work.

Additional information

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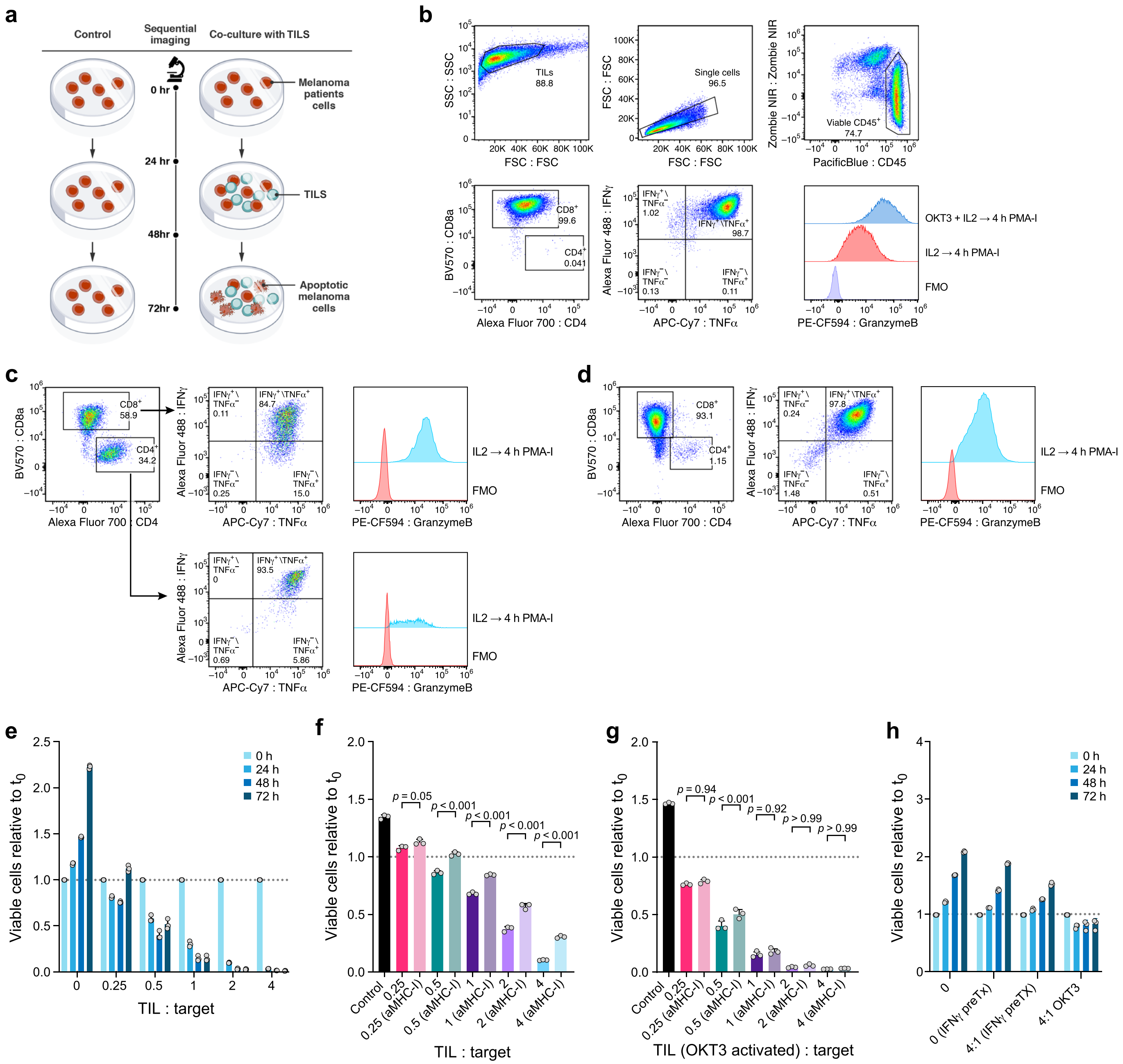
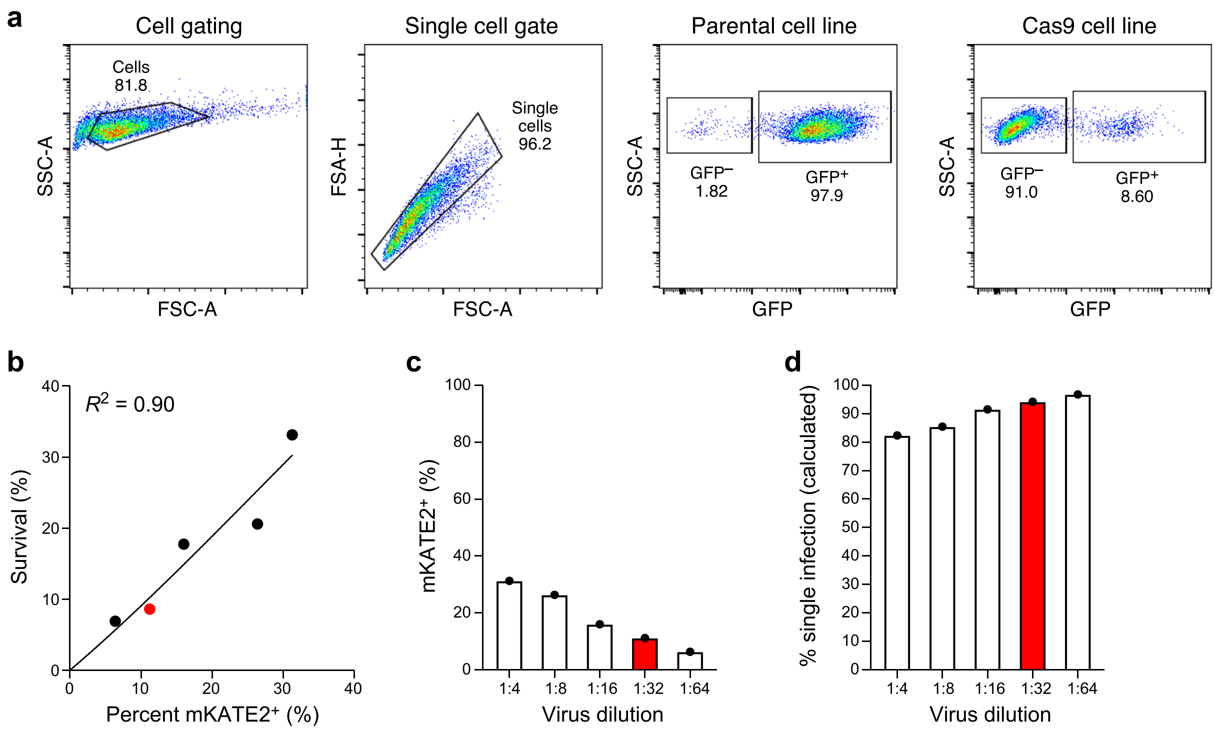
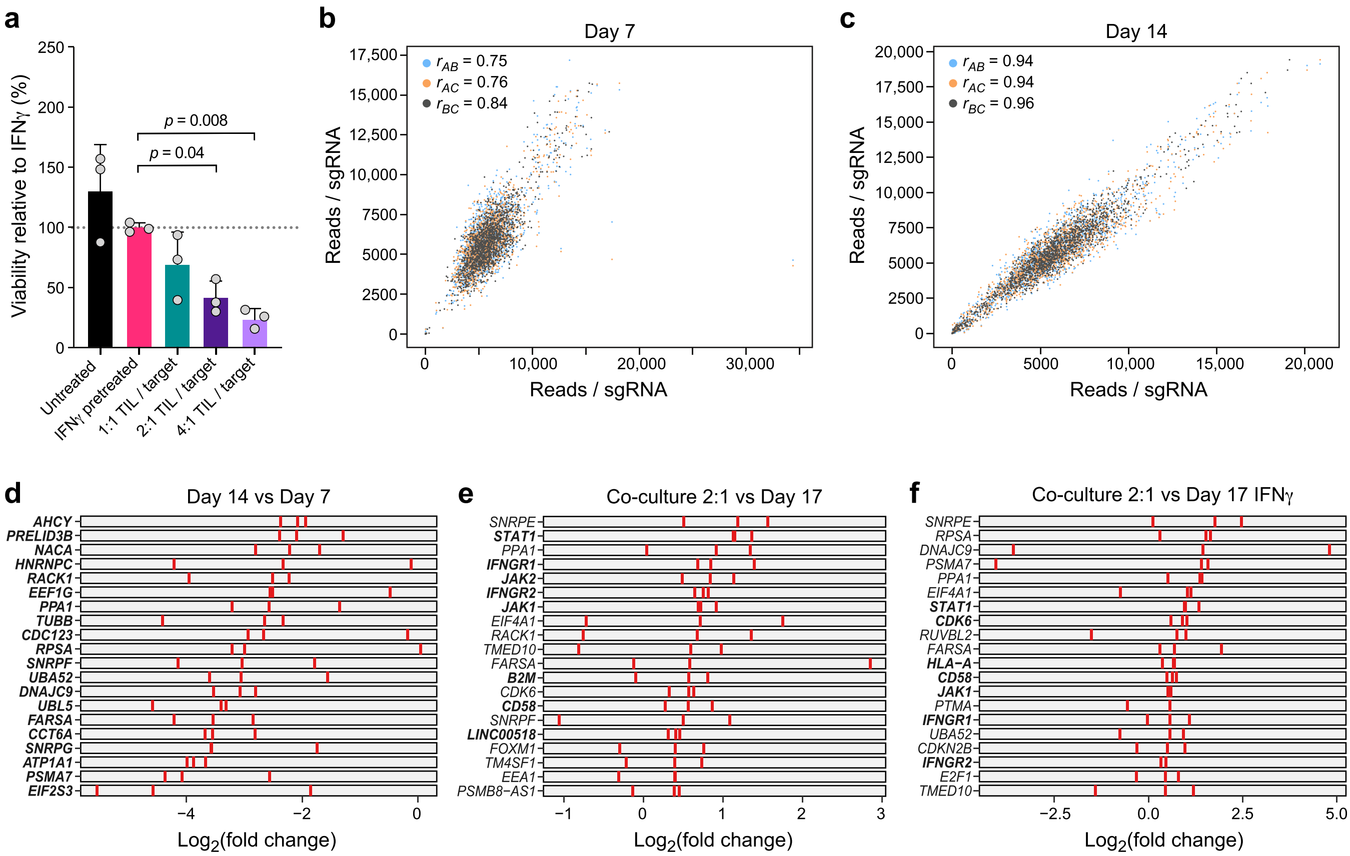


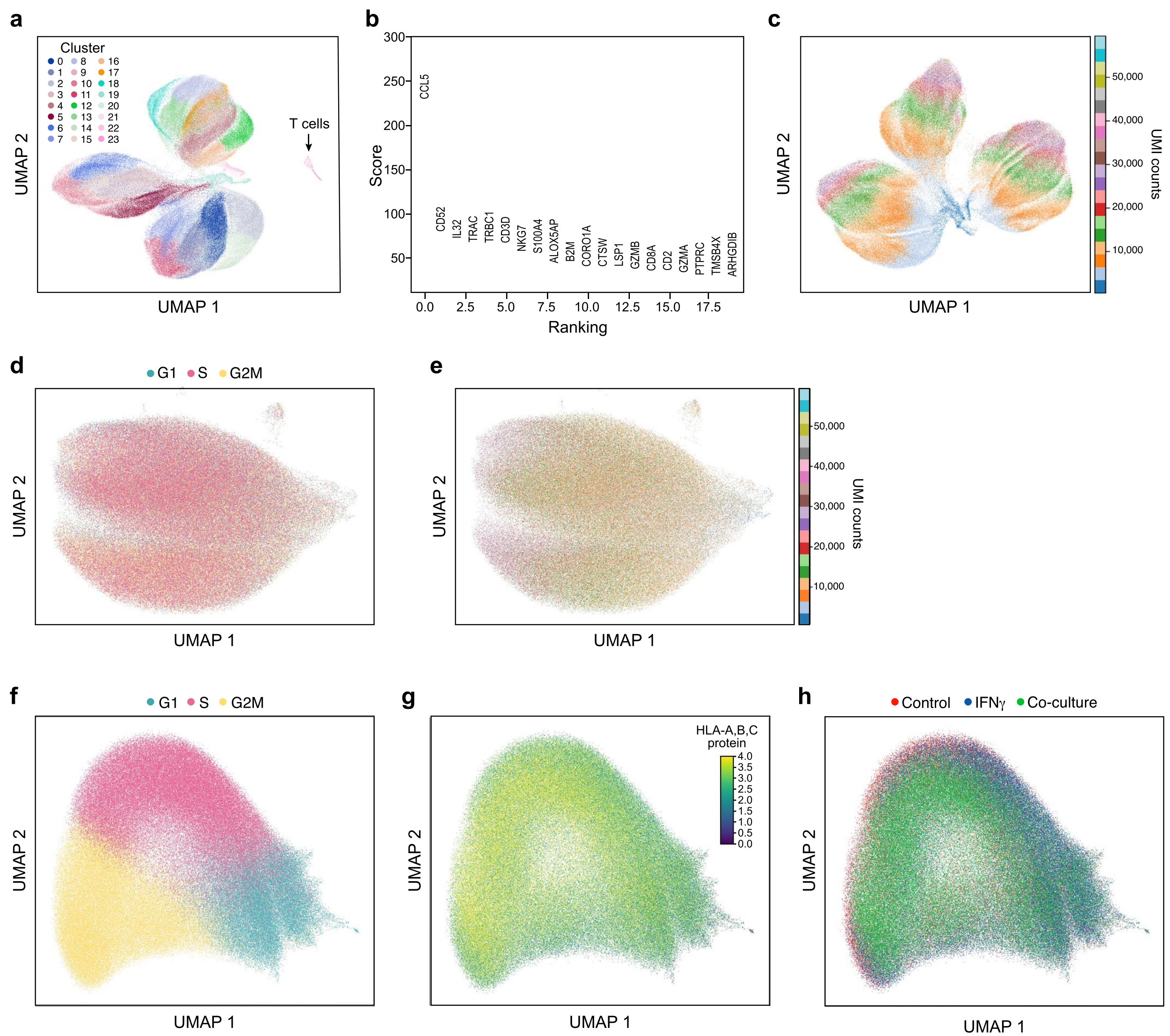
Extended Data Fig. 1 | Establishment of patient derived co-culture model. a, Approach for imaging-based quantification of TIL-mediated killing of melanoma target cells. Plates were imaged at 0, 24, 48 and 72 hours, and viable cell counts were normalized to starting counts to quantify outgrowth of target cells. **b**-**d**, Sorting and gating strategy to isolate and expand TIL cultures prior to co-culture. b. TILs grown in IL2 or OKT3-stimulated for 72 hours and analyzed after 4 hours of PMA-I-stimulation. FMO, fluorescent-minus-one control b. TILs from 2686 retain ability to induce IFNγ and TNFa, and OKT-3 reactivation leads to an increase in Granzyme-B production compared to TILs grown in IL2 alone. **c**, MaMel-134 TILs produce IFNγ, TNFa, and Granzyme-B. **d**, MaMel-80 TILs produce IFNγ, TNFa, and Granzyme-B. e-h. Impact of time, dose, IFNγ pre-treatment, MHC-I blocking, and OKT3 on TIL-mediated killing in the co-culture system from patient 2686. Ratio of viable cancer cells (y axis, relative to t0) in co-cultures: (**e**) after different time points of co-culture at increasing TIL:cancer cell ratios (x axis), where TILs were restimulated with immobilized OKT3 for 72 h prior to co-culture; **f**, after 48 h of co-culture, where cancer cells were pre-treated with 1 ng/ml IFNγ for 16 hours (without prior OKT3-reactivation); **g**, after 48 hours of co-culture as in (**f**) but using OKT3-reactivated TILs. **h**, Specificity of IFNγ pre-treatment approach. Ratio of viable allogenic cancer cells (y axis, relative to t0) in different culture conditions with or without IFNγ pre-treatment (x axis) grown from 0 to 72 hours (color bars) with 2686 TILs with or without prior reactivation with OKT3. For **e**-**h**, we performed a one-way ANOVA with Tukey post hoc test. Error bars: Mean±SD. All experiments were performed in triplicates in each of at least two independent experiments.



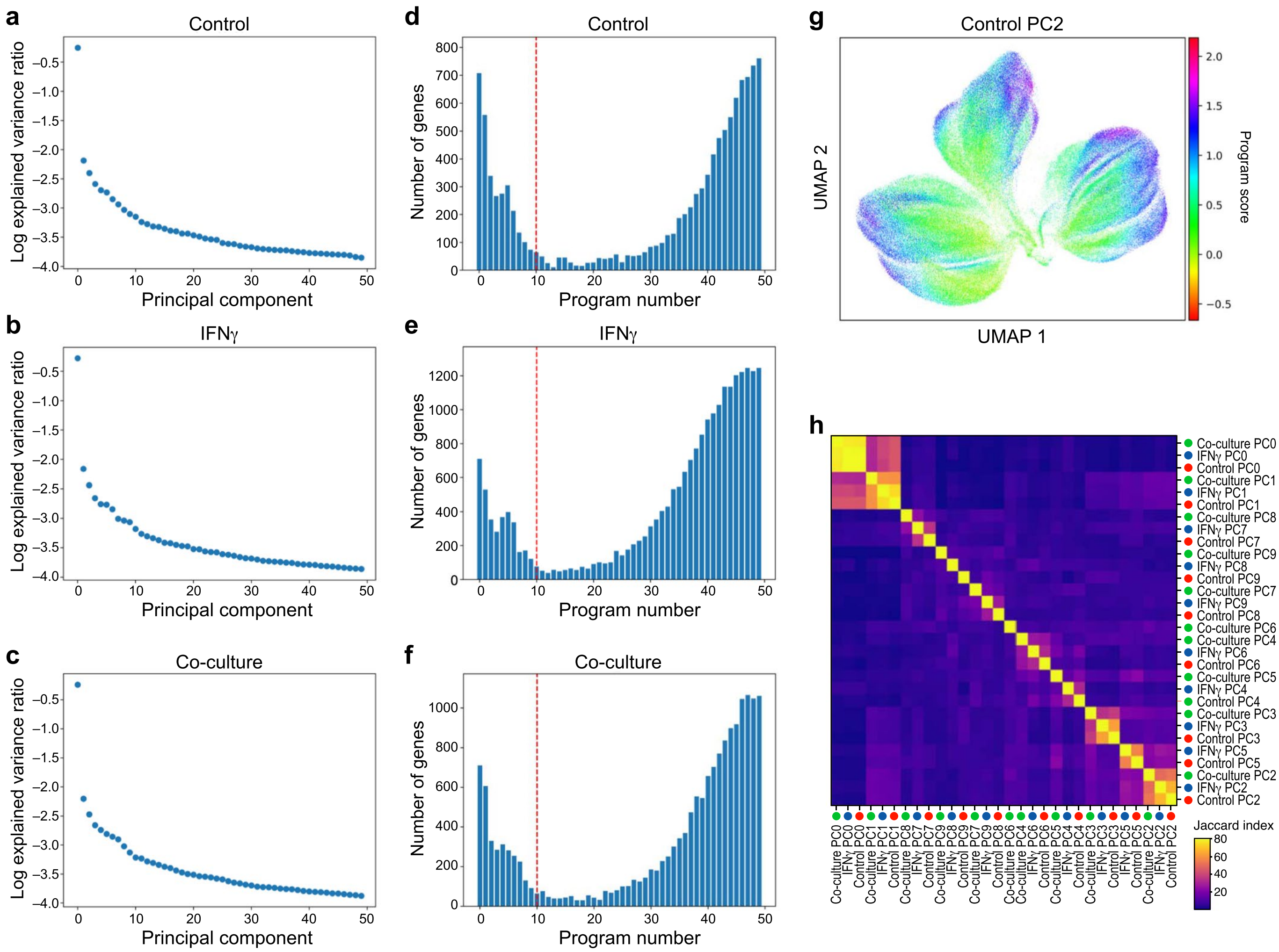
Extended Data Fig. 2 | Generation of Cas9 transgenic patient derived lines and sgDNA library titration. a, High Cas9 activity in Cas9 transgenic line. Flow cytometry of EGFP levels in Cas9-expressing and parental melanoma cells from patient 2686 transduced with lentivirus encoding EGFP and an EGFP targeting sgRNA at MOI <1 and selected using puromycin. **b-d**, Transduction of sgDNA lentiviral library to Cas9 transgenic line. **b**, Proportion of mKATE2+ cells prior to selection (x axis) and survival after puromycin selection (y axis) in 2686 melanoma Cas9 transgenic cells transduced with the ICR library. Line: Linear regression, Pearson R2=0.90. **c**, Percentage of mKate2+ cells (y axis) in 2686 melanoma Cas9 transgenic cells transduced with the ICR library at virus dilutions (x axis). Red: Dilution used for the Perturb-CITE-seq screen. d. Proportion of cells estimated to be infected by one virus (y axis) at different dilutions of the ICR library (x axis). Red: Dilution used for the Perturb-CITE-seq screen.



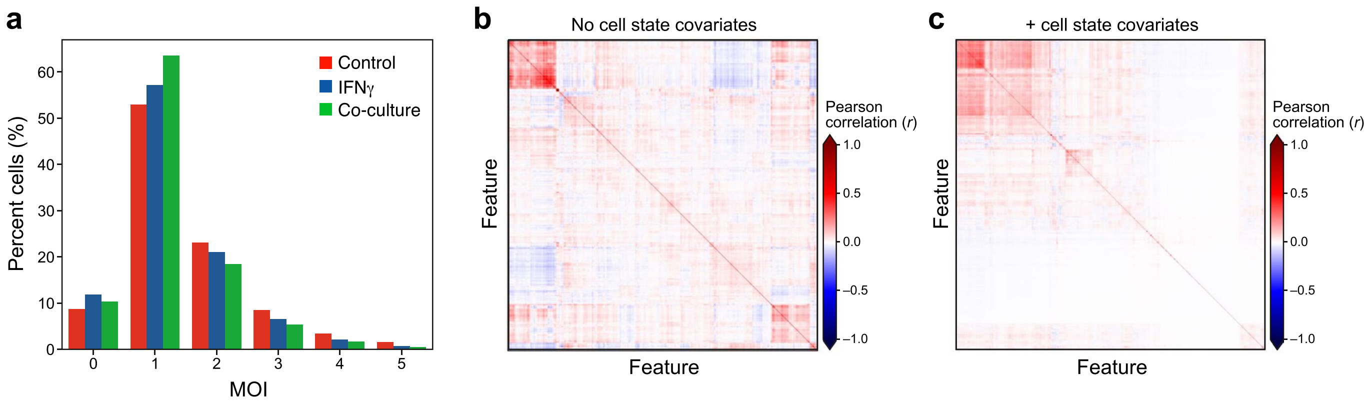
Extended Data Fig. 3 | CRISPR/Cas9 viability screen in the co-culture system. a, Dose-response killing in the co-culture experiment validates target killing range. Percent of surviving cells relative to IFNγ pretreated target cells (y axis) in different co-culture conditions (x axis) from a plate run in parallel to the viability and Perturb-CITE screens, with triplicate wells for each condition. One-way ANOVA with Dunnet post-hoc test. Error bars: Mean±SD. **b**,**c**, Screen reproducibility across triplicates. Number of reads detected (x, y axis) for each sgDNA (dots) when comparing each pair within triplicate experiments (color legend) in pre-treated day 7 (**b**) or day 14 (**c**). Pearson correlation coefficients are noted in the color legend. **d-f**, Identification of essential genes and genes affecting resistance to TIL mediated killing. Relative depletion (log2(FC), x axis) for each individual sgDNA (red bar) of the top20 target genes by MAGeCK analysis (rows) (n = 3 sgDNAs/target gene, Methods) on day 7 (without TILs, to recover essential genes, **d**), day 17 of 2:1 TIL:cancer cells co-cultures, comparing control cells (**e**), or day 17 of IFNγ treated cells (**f**, no co-culture). Bold lettering indicates significantly enriched / depleted target.



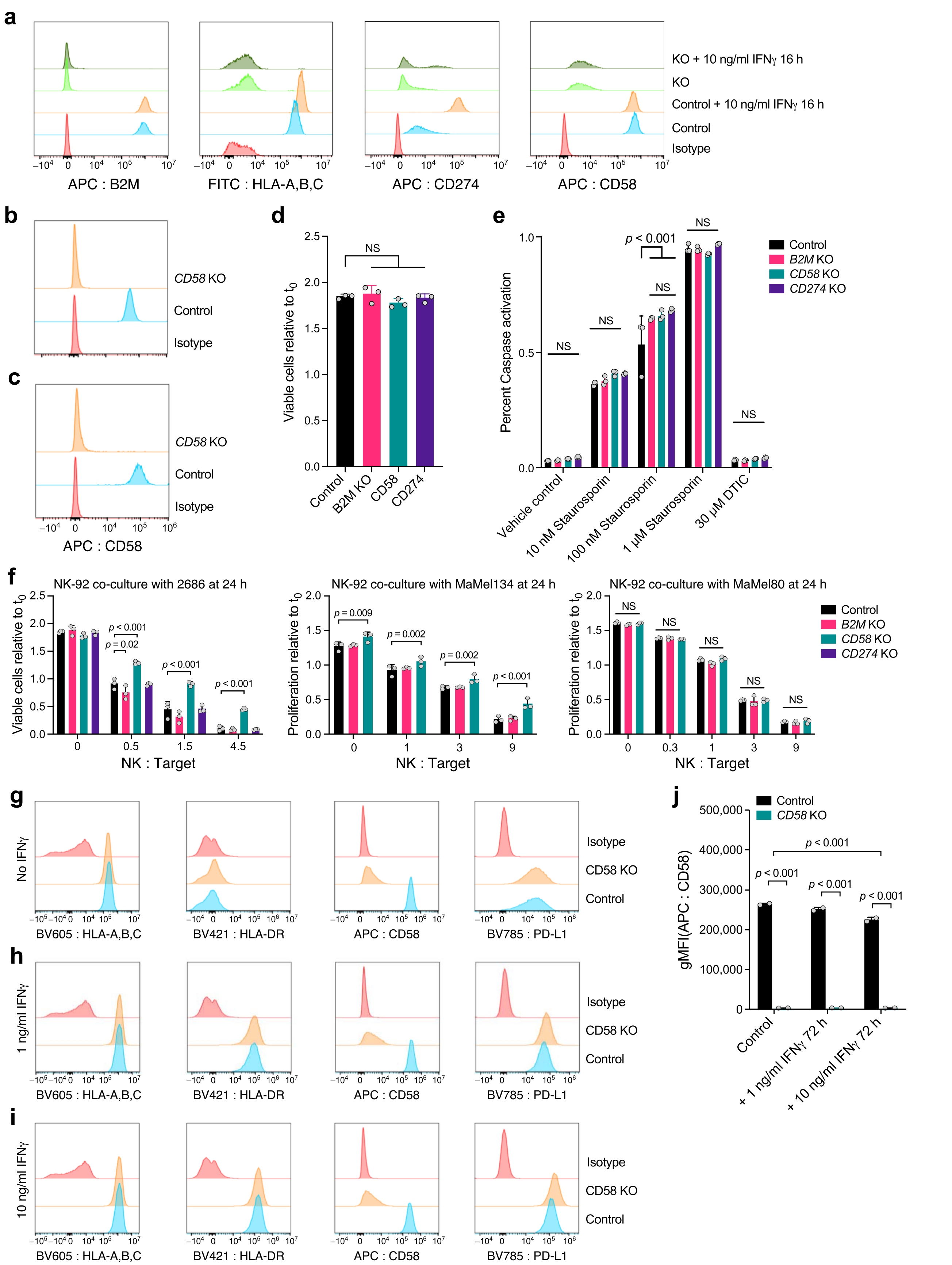
Extended Data Fig. 4 | Characterization of different immune pressures by single-cell RNA and protein profiles. a,**b**, Removal of profiled T cells. a. UMAP embedding of single cell RNA-Seq profiles from the Perturb-CITE-seq screen, colored by unsupervised cluster assignment[34] (Methods). **b**, A permutation test was used to score marker genes associated with each cluster shown in (**a**)[35]. Score (y axis, permutation test, Methods) of marker genes (x axis) associated with the distinct cluster marked by an arrow in (**a**), include canonical T cell markers. **c**, Cell complexity affects scRNA-Seq profiles. UMAP embedding of scRNA-Seq profiles of cancer cells only, colored by UMI count bins. **d**,**e**, CITE profiles of 20 cell surface proteins do not reflect cell cycle phases (**d**) or UMI count (**e**). UMAP embedding of cells (dots) by CITE-seq profiles (dots) colored by cell cycle phase, as scored from scRNA-Seq of the cells, and (**e**) UMAP of cells by count bins (indicated in legend). **f-h**, Limited relation between the cell cycle and immune pressure or phenotype. UMAP embedding of cells (dots) based only on RNA expression on cell cycle genes colored by (**f**) cell cycle phase based on the cell's RNA profile; (**g**) MHC protein levels from the CITE signal of the cell; or (**h**) condition.



Extended Data Fig. 5 | Learning expression programs in different conditions. a-f, Identification of programs by jackstraw PCA in each condition. a-c. Explained variance (y axis) by each principal component (x axis) for PCA performed on control (**a**), IFNγ-treated (**b**), or co-culture (**c**) Perturb-CITE-seq data. **d-f**, Number of features (y axis) for each jackstraw program (x axis) for models learned on control (**a**), IFNγ-treated (**b**), or co-culture (**c**) Perturb-CITE-seq data. Dotted red line: cutoff for programs considered in further analysis. **g**, G2M program learned from control dataset. UMAP embedding of cells (dots) by scRNA-seq profiles, with cells colored by the gene set score (color bar) (Methods) of a G2M cell cycle control program (compare to Fig. 3f) (Methods). **h**, Identifying related programs across conditions. Jaccard index (color bar) for each pair of programs across all 30 programs (rows).

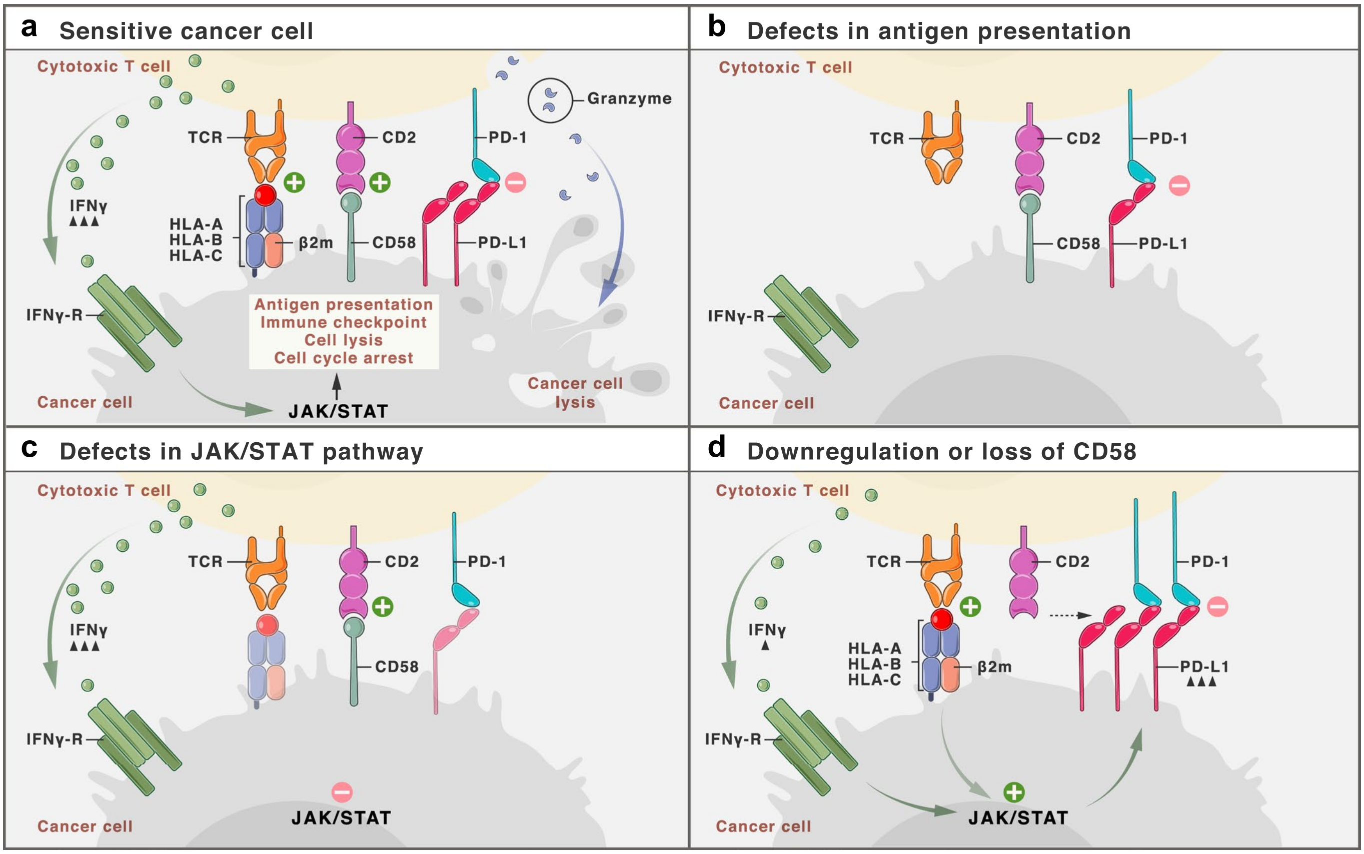


Extended Data Fig. 6 | Addressing cell cycle and complexity covariates by the Perturb-CITE-seq model and impact of targeting vs. non-targeting guides. **a**, Estimated Multiplicity of infection (MOI). Distribution of cells (%, y axis) at different estimated MOI (x axis) in each experimental condition (color legend) as determined from the guide dictionary (Methods). **b**,**c**, Improved model fit following accounting for cell state as a covariate. Pearson correlation (color bar) between the residuals from the linear model fit for each regulated feature (columns) from models learned without (**b**) or with (**c**) cell state covariates accounting for the cell cycle and cell complexity.



Extended Data Fig. 7 | See next page for caption.

Extended Data Fig. 7 | Role of CD58 in resistance to T cell mediated killing and regulation of PD-L1. a, Validation of CRISPR/Cas9 KO (and unperturbed control) melanoma cells. **a**, Distribution of fluorescent intensity of CD58 (APC-CD58), B2M (APC-B2M), MHC-I (FITC-HLA-A,B,C), PD-L1 (APC-CD274) or staining with isotype control, without or with IFNg stimulation, in 2686 melanoma cells. **b**, Validation of *CD58* KO MaMel-134, and **c**, MaMel-80 melanoma cells. **d**, Comparable growth of control and KO cells. Ratio of viable cells relative to timepoint 0 (y axis) for control and *B2M* KO, *CD58* KO or *CD274* KO melanoma cells from patient 2686 (x axis). **e**, Comparable induction of apoptosis in response to Staurosporin and resistance to DTIC in control and KO melanoma cells. Percent of cells inducing Caspase 3/7 (y axis) in control and *B2M* KO, *CD58* KO or *CD274* KO melanoma cells (color code) from patient 2686 in different treatment conditions (x axis). **f**, Co-culture experiments of three melanoma cells lines (2686, MaMel134 and MaMel80) at increasing ratios of NK cells and tumor cells with different genotypes (including *CD58* KO, *B2M* KO in all models, and additionally *CD274* KO in 2686). **g-j**, *CD58* perturbation in co-culture does not affect *B2M* and *HLA* expression at the RNA and protein level but induces *CD274*. Distribution of fluorescent intensity by flow cytometry (corresponding to Fig. 5i-k) of MHC Class I and II, CD58, and PD-L1 in parental (control) and *CD58* KO lines at baseline (**g**) and after 72 hours of stimulation with either 1 ng IFNγ (**h**) or 10 ng IFNγ (**i**), and summary of the impact of IFNγ on CD58 protein abundance after 72 hours. In all experiments, we used a one-way ANOVA with Tukey post hoc test. Error bars: Mean ±SD. All experiments were performed in triplicates in each of at least two independent experiments.



Extended Data Fig. 8 | Known mechanisms of immune evasion and distinct role of CD58. a, Schematic representation of interactions between T cell and cancer cells. At baseline, TCR stimulation via peptide-loaded MHC Class I and through CD58:CD2 co-stimulation results in production of cytokines (such as IFNγ, Granzymes), which lead to activation of the IFNγ-JAK/STAT-pathway that determine the cell fate and expression of surface proteins. **b**, In the absence of antigen presentation, either due to genetic defects or downregulation of the antigen presentation machinery, cancer cells survive. **c**, Defects in the JAK/STAT-pathway (such as deleterious mutations in JAK genes) result in poor response of cancer cells to T cell secreted IFNγ and promote cancer cell survival. **d**, Loss or downregulation of CD58 does not interfere with antigen presentation or JAK/STAT-signaling per se, but reduces T cell co-stimulation while simultaneously promoting increased co-inhibitory signaling via PD-L1, and therefore overall confers resistance to T cell mediated killing.

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Imaging data of co-cultures was acquired with Nexcelcom Celigo Imaging and analyzed with corresponding software package v.4 .1.3.0. Flow cytometry data was acquired using Sony Spectral Cytometer Software 1.6.3.7151 or Cytek Aurora Spectral Analyzer Software SpectroFlo v2.1.
Data analysis	Imaging data of cocultures was analysed with Nexcelcom Celigo Imaging softwarec,o ntrol normalized using Microsoft Excel 365. Plotting and statistical analysis was performed using Graphpad Prism 8.0. Flow cytometry data was analysed using Flow Jo 10. Plotting and statistical analysis was performed using Graphpad Prism 8.0.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample size for co-culture time-course and validation studies and sample size for flow cytometry was determined in pilot studies and is sufficient for rigorous statistical comparisons.
Data exclusions	No data was excluded.
Replication	All functional experiments were performed at least twice (biological replicates) with multiple technical replicates.
Randomization	No randomization was required, because we did not test activity of perturbations in a blinded fashion.
Blinding	No investigator blinding was performed, because treatments (i.e. TILs) had to be matched to the autologous cancer cell, and furthermore, blinding was not possible for other experiments (e.g. flow-cytometry) because specific perturbations results in differences in fluorescecne signal. Nonetheless, all experiments were performed by two independent investigators with concordant results.

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Materials & experimental systems

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☒	☐	Palaeontology and archaeology
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☒	☐	Human research participants
☒	☐	Clinical data
☒	☐	Dual use research of concern

Methods

n/a	Involved in the study	
☒	☐	ChIP-seq
☐	☒	Flow cytometry
☒	☐	MRI-based neuroimaging

Antibodies

Antibodies used	Flow Cytometry: (clone, manufacturer, catalog ID: gdTCR-BV421 (Bl, Biolegend, 331218), dilution 1:20, CD45-Pacific Blue (HI30, Biolegend, 304022), dilution 1:40, CD3-AF532 (eBiosciences, 58/0038-42), dilution 1:40, CD8a-BV570 (RPA-T8, Biolegend, 301038), dilution 1:40, Granzyme B-PE-CF594 (GBll, BD, 562462), dilution 1:50, FoxP3-APC (236A/E7, eBiosciences, 17-4777-42), dilution 1:20, CD4-AF700 (RPA-T4, Biolegend, 4345826), dilution 1:20, IFNy-AF488 (4S.B3, Biolegend, 502517), dilution 1:20, IL-17A-PE-Cy7 (BL168, Biolegend, 512315), dilution 1:20, TNF-a-APC-Cy7 (Mabll, Biolegend, 502943), dilution 1:40, B2M-APC (2M2, Biolegen, 316312), dilution 1:100, CD58-APC (TS2/9, Biolegend, 330918), dilution 1:100, CD274-APC (29E.2A3, Biolegend, 329708), dilution 1:100, HLA-A,B,C-FITC {W6/32, 311403, Biolegend), dilution 1:100, HLA-DR-B V421 (L243, Biolegend, 307636), dilution 1:100, HLA-A,B ,C-B V605 (W6/32, Biolegend, 311431), dilution 1:100, CD274-BV785 (29E.2A3, Biolegend, 329735), dilution 1:100, mouse IgG1, κ Isotype Ctrl (FC)-APC (MOPC-21, Biolegend, 400121), dilution 1:100, mouse IgG2b, κ Isotype Ctrl-APC (MPC-11, Biolegend, 400321), dilution 1:100.CITE-Seq (all antibodies used at 1:500 dilution unless indicated otherwise): Human TruStain FcX™ (Fc Receptor Blocking Solution) (Biolegend, 422301), dilution 1:20, CD117 (104D2, Biolegend, TotalSeq A #61), CD119 (GIR-208, Biolegend, TotalSeq A #219), CD140A (16A1, Biolegend, TotalSeq A#128), CD140B (18A2, Biolegend, TotalSeq A #129), CD172a (15-414, Biolegend, TotalSeq A #408), CD184 (12G5, Biolegend, TotalSeq A #366), CD202B (33.1, Biolegend, TotalSeq A #588), CD274 (29.E2.A3, Biolegend, TotalSeq A #7), CD279 (EH12.2H7, Biolegend, TotalSeq A #88), CD29 (TS2/16, Biolegend, TotalSeq A #369), CD309 (7D4-6, Biolegend, TotalSeq A#362), CD44 (BJ18, Biolegend, TotalSeq A #125), CD47 (CC2C6, Biolegend, TotalSeq A #26), CD49f (goH3, Biolegend, TotalSeq A #70), CD58 (TS2/9, Biolegend, TotalSeq A #174), CD59 (p282(H19), Biolegend, TotalSeq A #361), CD61 (VI-PL2, Biolegend, TotalSeq A #372), CD9 (HI9a, Biolegend, TotalSeq A #579), HLA-ABC (W6/32, Biolegend, TotalSeq A #58), HLA-E (3D12, Biolegend, TotalSeq A #918), Mouse IgG1 isotype Ctrl (Biolegend, TotalSeq A #90), Mouse IgG2a isotype Ctrl

	(Biolegend, TotalSeq A #91), Mouse IgG2b Isotype Ctrl (Biolegend, TotalSeq A #92), Rat IgG2a Isotype Ctrl (Biolegend, A #238). anti human HLA-ABC functional grade (W6/32, eBioscience, 16-9983-38), dilution 1:20
Validation	All fluorescently conjugated antibodies were validated for the specific application by the manufacturer and validation data is available on the manufacturer's website. Fluorescent-minus-one (FMO) staining controls were prepared for gating of intracellular antigens (IFN-g, TNF-a, IL-17A, Granzyme B). Isotype controls were used for gating negative populations of CD58, CD274 and HLA-ABC for FACS. Anti human HLA-ABC antibody used in cell culture studies was purified by affinity chromatography by the manufacturer and validated by the manufacturer for purity and endotoxin levels. This antibody clone has been described in Barnstable et al, Cell. 1978;14:9–20. doi: 10.1016/0092-8674(78)90296-9, and used as a blocking antibody for HLA specific cytotoxic interactions of CD8+ T cells in Ware et al, Immunity, 1995 Feb;2(2):177-84. doi: 10.1016/s1074-7613(95)80066-2 among others.

Eukaryotic cell lines

Policy information about cell lines

Cell line source(s)	Cell line source(s) Common cell lines: HEK-293T (ATCC) Patient derived cell lines: Melanoma cell line 2686 and matched tumor infiltrating lymphocytes (Tl Ls) were previously derived (under IRB protocol #2004-0069) and can be obtained from MDACC, Texas, USA. Melanoma cell lines Ma-Mel90 and Ma-Mel-134 and can be obtained from UK-Essen, Germany.
Authentication	None
Mycoplasma contamination	All cell lines were repeatedly tested negative for Mycoplasma and other bacterial contaminants using PlasmoText (InvivoGen, San Diego, CA, USA)
Commonly misidentified lines (See ICLAC register)	None used in this study.

Flow Cytometry

Plots

Confirm that:

☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☒ All plots are contour plots with outliers or pseudocolor plots.

☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	All samples were prepared from in vitro cultures. Adherent cells were lifted using Accutase. Accutase was quenched with media containing 10$ FBS and cells were subjected to further staining procedures as outlined in the Methods section of the manuscript.
Instrument	Analysis: Sony SP6800 and Cytek Aurora 4-laser system.
Software	FlowJo, v.10.
Cell population abundance	Cell population abundance for the Perturb-CITE-seq experiment after co-culture was ~30% of all cells. In all other experiments, we used pure cancer cell line cultures, therefore all cells were cancer cells without contamination.
Gating strategy	First, cells were gated by FSC-A vs. SSC-A, then single-cells were gated using FSC-A vs. FSC-H. Viable cells were identified using ZOMBI-NIR-negative cells and used for surface marker analysis (tumor cells) or gated on CD4 vs. CD* (TILs) and used to quantify intracellular cytokine levels. Gates for intracellular cytokines were set based on fluorscent-minus-one (FMO) controls.

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.